

Name: _____

AS Biology Unit 1

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2016					
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Boundaries

Year	E	D	C	B	A	Max UMS
2016	22	27	32	37	42	52
2017	17	22	28	34	40	52
2018	20	26	32	38	44	56
2019	19	24	30	36	42	54
2022	11	17	23	29	35	47
2023	16	22	28	35	42	56



Surname	Centre Number	Candidate Number
Other Names		2



GCE AS/A Level

2400U10-1 – **NEW AS**



BIOLOGY – Unit 1
Basic Biochemistry and Cell Organisation

P.M. THURSDAY, 26 May 2016
1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	8	
2.	16	
3.	11	
4.	19	
5.	17	
6.	9	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the continuation pages at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

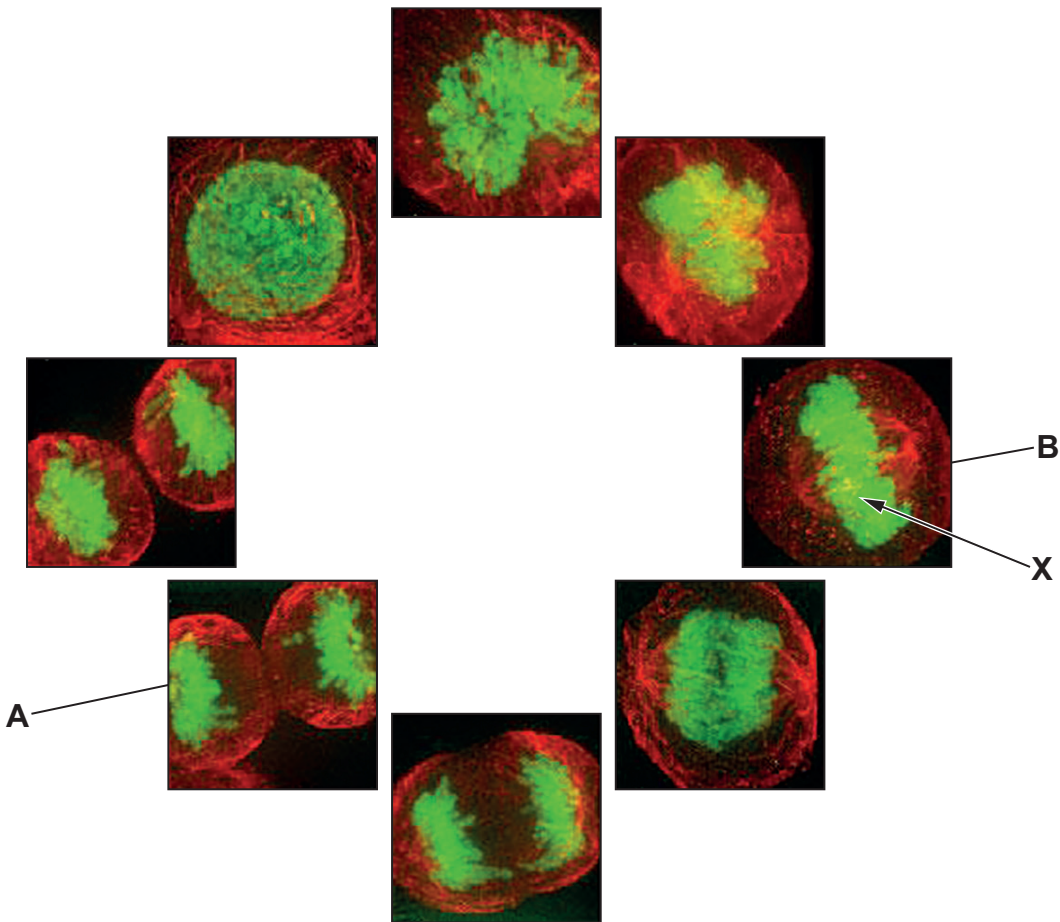
The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question 6.



MAY162400U10101

Answer all questions.

1. In 1951, Henrietta Lacks died of cervical cancer. A research scientist called George Gey, grew a sample of her tumour. He found that these cells multiplied rapidly and could be grown indefinitely in culture. They became the first immortal human cell line, which he named HeLa. HeLa cells are now grown in research laboratories in many different countries.
- (a) The photomicrographs below show images of HeLa cells during different stages of the cell cycle. The cells have been stained with a dye, which causes the DNA to be visible.



- (i) Name the stages shown in **A** and **B**.

[2]

Stage **A**:

Stage **B**:



Examiner
only

(ii) Name the structures, labelled X, that can be seen in the photomicrograph. [1]

.....

(iii) Explain why some stain would be seen in other parts of the cell. [2]

.....

.....

.....

.....

(b) A biotechnology company that supplies HeLa cells to laboratories states that the number of HeLa cells double every 19 hours. They suggest that the starting culture of cells should have a density of 30 000 cells cm⁻³ and that the cells should be sub-cultured every 4 days.

Calculate the density of the cells that would be present after 4 days; give your answer in cells cm⁻³ in standard form. [3]

Cell density = cm⁻³

8

2400U101
03



Examiner
only

2. Influenza, commonly known as “the flu”, is an infectious disease caused by the influenza virus, which infects the epithelial cells of the upper respiratory tract. The virus is composed of a protein capsid which surrounds the enzyme, RNA polymerase and its genetic material, RNA.

(a) Describe the differences between the genetic material of the Influenza virus and the genetic material found within the nucleus of the epithelial cell it infects. [4]

.....

.....

.....

.....

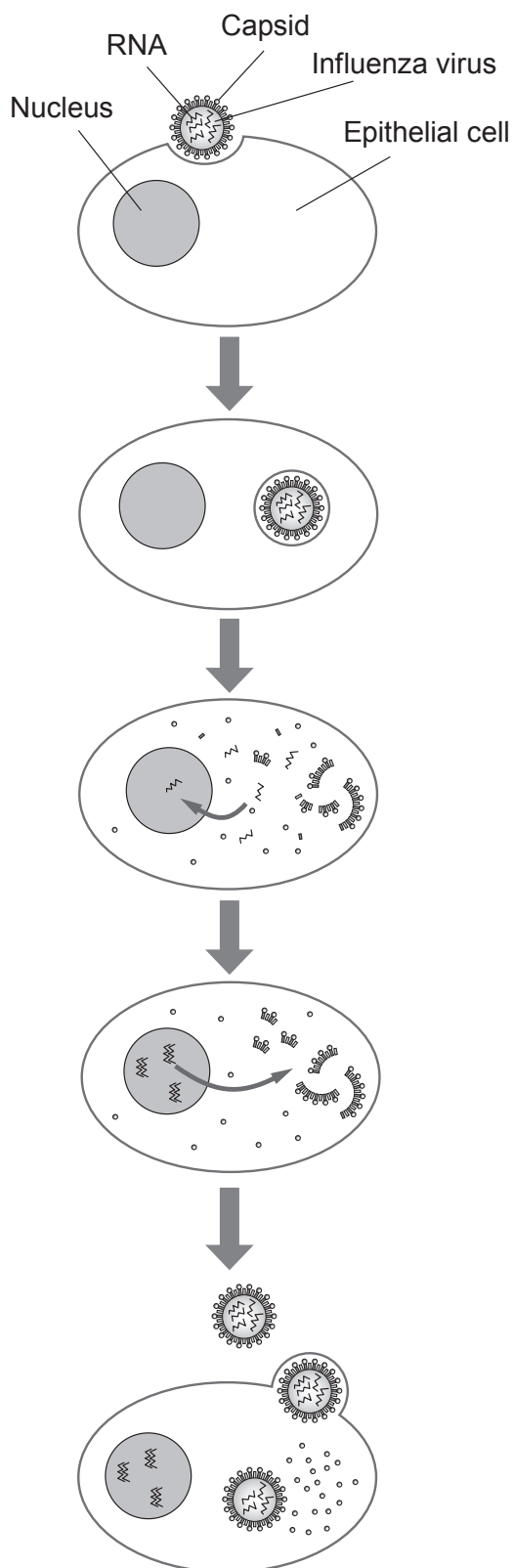
.....

.....

2400U101
05



- (b) The virus can only replicate in living (host) cells, where it utilises nutrients and organelles within the cell to multiply quickly. The diagram below shows the stages in the replication of the virus.



- 1 Influenza virus becomes attached to a target epithelial cell.
- 2 The cell engulfs the virus.
- 3 Viral contents are released. Viral RNA enters the nucleus where it is replicated by the viral RNA polymerase.
- 4 Viral mRNA is used to make viral proteins.
- 5 New viral particles are made and released into the extracellular fluid. The cell, which is not killed in the process, continues to make a new virus.



(i) State the name of the process occurring between stages 1 and 2. [1]

(ii) In stage 3, the capsid has broken down and the viral RNA and RNA polymerase enter the nucleus, where the viral RNA is replicated. Apply your knowledge of RNA polymerase to describe how the viral RNA is replicated. [4]

(iii) To complete the replication of the virus, capsids need to be produced (stage 4). Describe how the proteins in the capsid are produced. [5]



(iv) Explain why the virus is unable to synthesise its own capsid.

[2]

.....

.....

.....

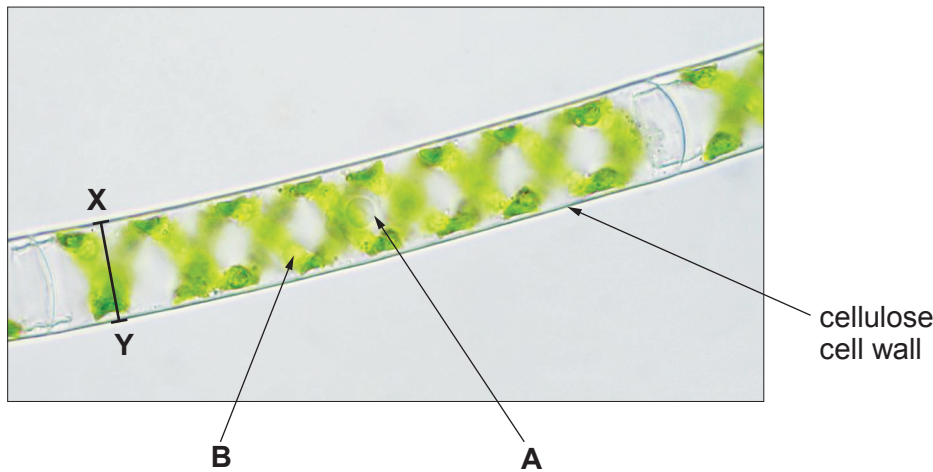
.....

Examiner
only

16



3. The photomicrograph below is of *Spirogyra*; an autotrophic organism that inhabits fresh water ponds and ditches.



(a) Identify the organelles labelled **A** and **B**. [2]

A:

B:

(b) The actual width of the cell between points **X** – **Y** was 32.3 μm . Calculate the magnification that was used to take the photomicrograph. [2]

magnification =

(c) During the day the concentration of solutes in the pond water changes due to evaporation. However, the length of the *spirogyra* cells remain almost constant. What can you conclude about how the structure of the cell wall enables *Spirogyra* to survive in different solute concentrations? [4]

.....

.....

.....

.....

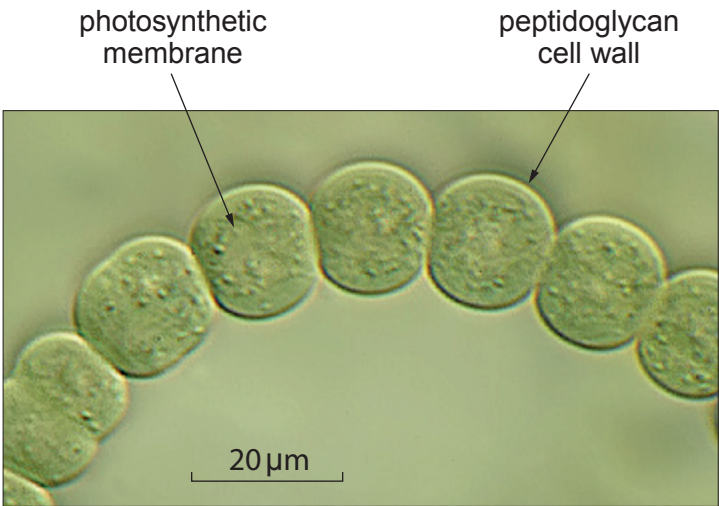
.....

.....



(d) Below is a photomicrograph of the bacteria *Nostoc*, which also inhabits fresh water ponds and ditches. They were once thought to belong to the same group of organisms as *Spirogyra* and thought to have the same cellular structure. Evidence from electron microscopy has now grouped these two separately.

Conclude what cell types are present in *Spirogyra* and *Nostoc*. Identify **two** differences and **one** similarity (not labelled in the photomicrographs) between these two species that would be revealed by electron microscopy. [3]



.....

.....

.....

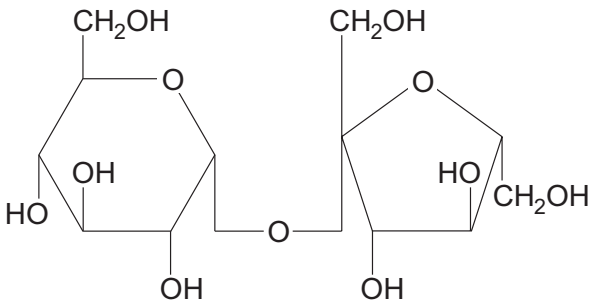
.....

11

2400U101
11



4. Sucrose is a disaccharide of glucose and fructose. The enzyme sucrase, catalyses the hydrolysis of sucrose into its monosaccharides. A colorimeter can be used to record absorbance values, which can then be used to determine the rate of hydrolysis.
- (a) (i) Complete the diagram to show the hydrolysis of sucrose including the products formed. [2]



(ii) State the name of the bond broken during the reaction. [1]

(iii) Explain why glucose and fructose are referred to as structural isomers. [1]

(iv) A student was provided with two beakers, one containing sucrose and the other containing sucrase. Describe **one** biochemical test that the student could have carried out to distinguish between the two solutions. [2]

(b) A student wanted to investigate the hypothesis that sucrase catalyses the hydrolysis of sucrose fastest at a neutral pH. She was provided with the following method:

- Add 5 cm³ of buffer solution to a test tube
- Add 5 cm³ of sucrose solution to the test tube
- Add 1 cm³ of sucrase solution to the test tube and mix
- After 20 minutes add 1 cm³ of dinitrosalicylic acid (DNS)
- Pipette 5 cm³ of the mixture into a cuvette and place into a colorimeter and record the absorbance of light passing through the solution.

Dinitrosalicylic acid (DNS) will react with monosaccharides to produce amino-nitrosalicylic acid (ANS). ANS causes a colour change to occur which can be detected by a colorimeter. The greater the concentration of ANS the greater the absorbance of light.

(i) State **two** additional variables that the student should control to ensure that the results recorded would be repeatable. [2]



(ii) The results are shown in the table below.

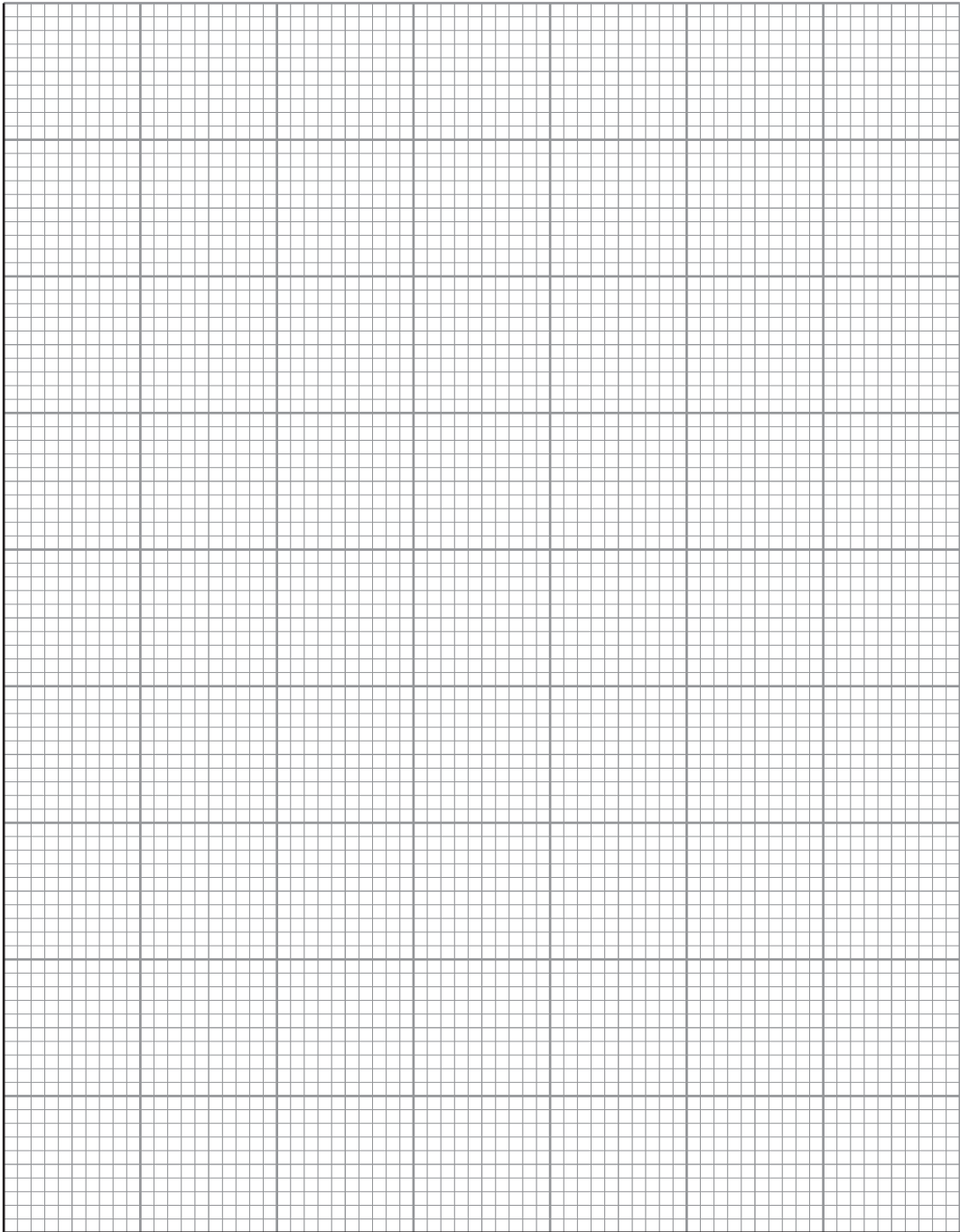
pH of sucrose and sucrase solution	Absorbance of light (au)			
	trial 1	trial 2	trial 3	mean
3	0.84	0.63	0.76	0.74
4	0.89	0.95	0.76	0.87
5	0.85	0.91	0.72	0.83
7	0.27	0.34	0.29	0.30
9	0.13	0.11	0.12	0.12
10	0.05	0.04	0.06	0.05



Plot the mean data on the graph below.

[4]

Examiner
only



(iii) What conclusions can be drawn from the data regarding the hypothesis being tested? [3]

.....

.....

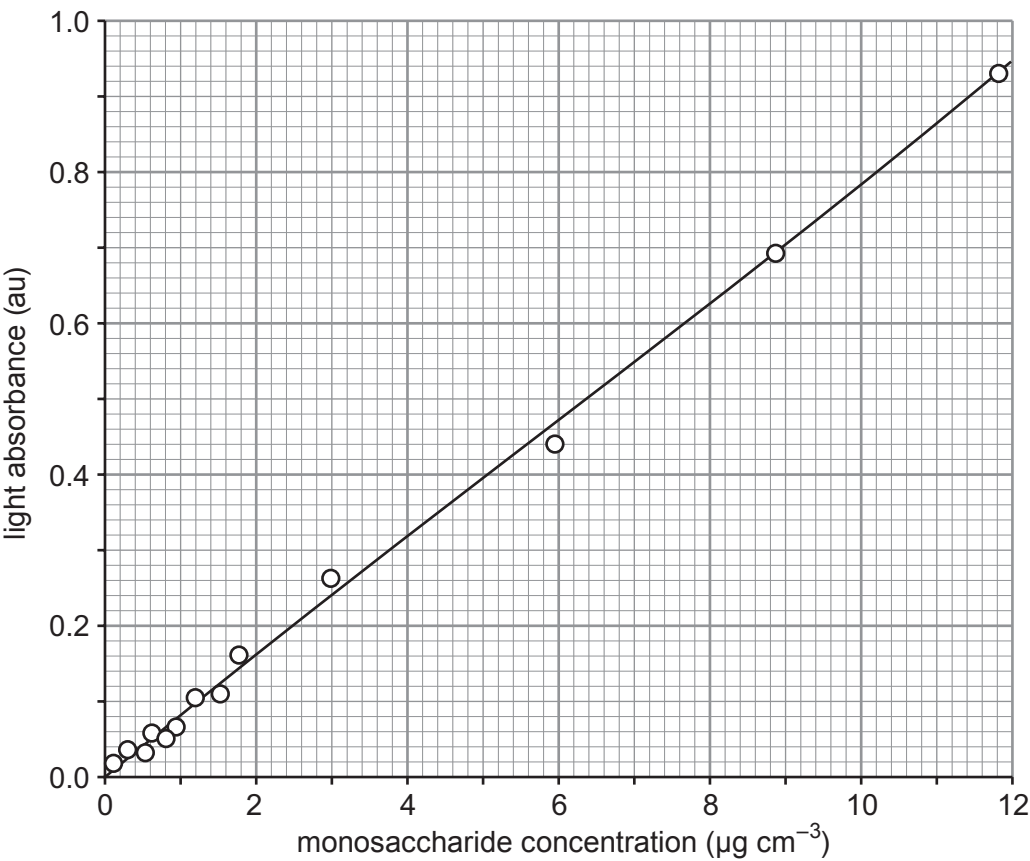
.....

.....

.....

(iv) The student was also provided with the calibration curve below in order to calculate the concentration of glucose produced during the investigation.

The colour change, caused by ANS, is dependent on the concentration of monosaccharides, the greater the concentration the greater the absorbance value recorded.



Determine the maximum concentration of glucose produced during the investigation at pH 7. [2]

.....



(c) With reference to the data explain how, and why, you would modify the method to determine the optimum pH more accurately. [2]

.....

.....

.....

.....

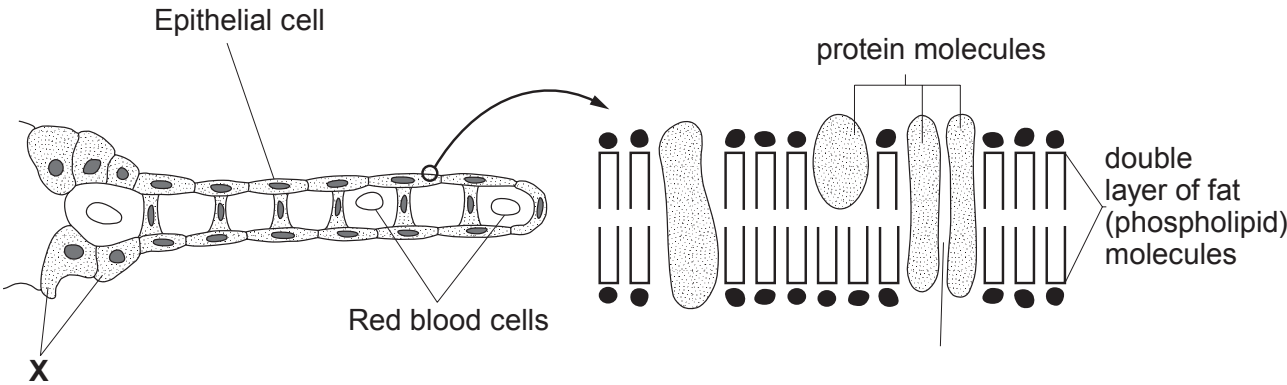
.....

Examiner
only

19



5. The primary role of the gills of fish is gas exchange. Oxygen, a non-polar molecule, passes from the surrounding water into the blood of the gills. The gills are also permeable to water and solutes, such as sodium ions and chloride ions.



- (a) Use the information provided to describe how oxygen and sodium ions cross the membrane. [4]

.....

.....

.....

.....

.....

.....



Examiner
only

(b) The data below shows the percentage ion composition of different aquatic habitats and of the blood plasma of two different fish.

		Percentage ion composition	
Fish	Habitat	Blood plasma of fish	Habitat
Flounder	Sea water	1.1	3.5
Carp	Fresh water	0.9	<0.1

Explain why it is important for the survival of the carp to produce large volumes of dilute urine. [4]

.....

.....

.....

.....

.....

.....

.....



- (c) (i) Specialised cells, labelled **X** on the diagram, are located in the epithelium of the gills. They are important in maintaining the ion composition of the blood plasma. Use the data to explain why this is necessary in the flounder. [3]

.....

.....

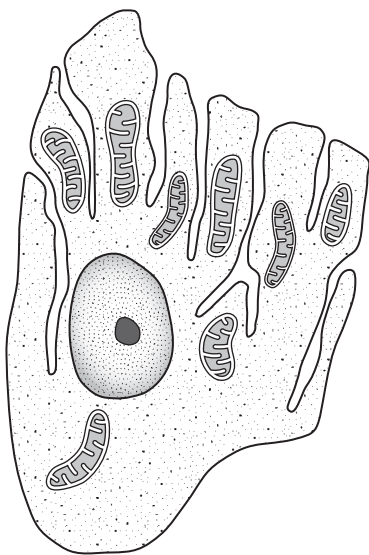
.....

.....

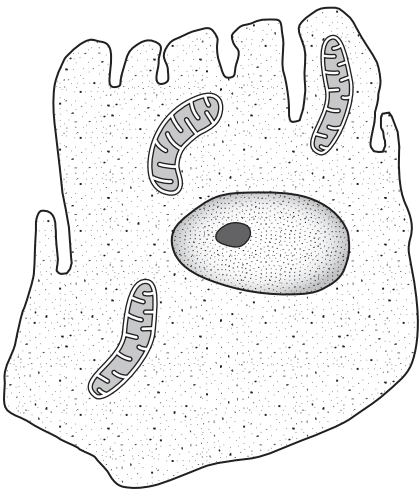
.....

- (ii) Salmon spend the majority of their lives in sea water. In order to reproduce salmon must migrate into fresh water rivers. During these migrations these specialised cells undergo structural changes.

The diagrams below are of these cells, taken from the salmon in the two different habitats. Giving reasons for your answer, suggest which cell was taken from the salmon in fresh water, and which was taken from the salmon whilst in sea water. [2]



Cell 1



Cell 2

.....

.....

.....

.....

.....



Examiner
only

(d) Acidification of fresh-water lakes, due to acid rain, has been linked to the death of fish such as carp. Scientists concluded that one of the causes of death in these animals is their inability to maintain blood plasma ion concentrations. Use the information to explain how they arrived at this conclusion. [4]

.....

.....

.....

.....

.....

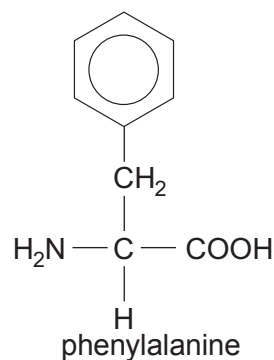
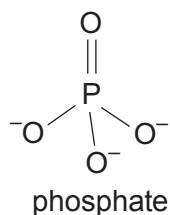
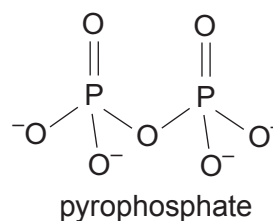
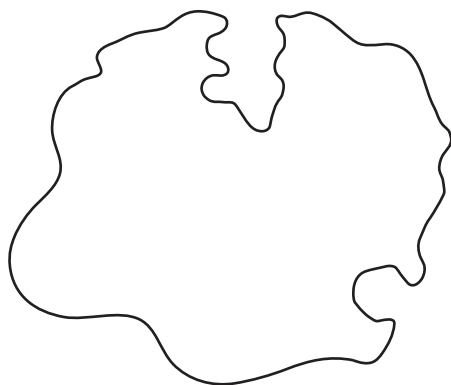
.....

.....

17



6. Pyrophosphatase is an enzyme found inside the nucleus of cells and is involved in DNA replication. The enzyme catalyses the conversion of a molecule of pyrophosphate to two phosphate ions. The diagrams below show the enzyme pyrophosphatase and its substrate pyrophosphate. Molecules of phenylalanine (an amino acid) and phosphate are also shown; both of these molecules are known to inhibit pyrophosphatase. (Drawings are not to the same scale).



Describe and explain why pyrophosphatase can only hydrolyse pyrophosphate and the mechanism by which phenylalanine and phosphate inhibit pyrophosphatase. [9 QER]



Examiner
only

Handwriting practice area with horizontal dotted lines.



Examiner
only



Examiner
only

9

END OF PAPER



Surname	Centre Number	Candidate Number
Other Names		2



GCE AS/A LEVEL – **NEW**

2400U10-1



BIOLOGY – AS unit 1
Basic Biochemistry and Cell Organisation

THURSDAY, 25 MAY 2017 – AFTERNOON

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	13	
3.	10	
4.	15	
5.	14	
6.	10	
7.	9	
Total	80	

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the continuation pages at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question 7.
The quality of written communication will affect the awarding of marks.



MAY172400U10101

Answer all questions.

1. The images below show a longitudinal section through a tissue found lining the trachea of humans and a highly magnified image of the surface of this tissue.

Image 1

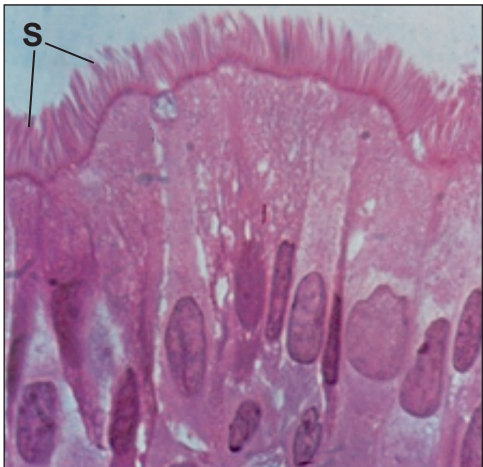
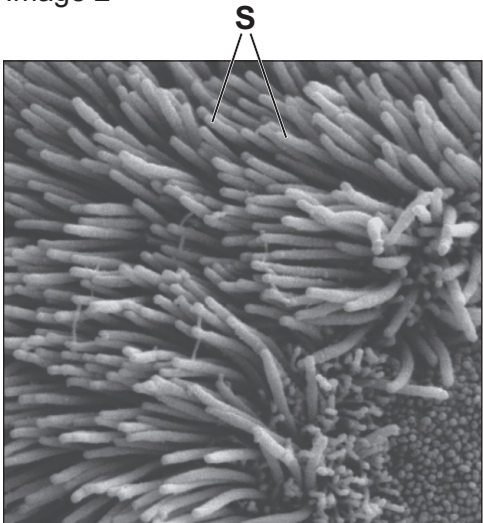


Image 2



- (a) (i) Name the structures labelled **S** and suggest their function in the human respiratory system. [2]

.....

.....

.....

- (ii) Explain why Image 1 is referred to as a tissue rather than an organ. [1]

.....

.....

.....



The photo electron micrograph below shows two mitochondria found in this tissue.



- (b) (i) One reason for the different appearance of the mitochondria is that they have been cut in different planes. Suggest **one other** reason that could account for the difference in their appearance. [1]

.....

.....

- (ii) Calculate the actual size of mitochondrion **T** along the line shown on the image. Give your answer in micrometres. [3]

Length of organelle = μm

- (iii) State and explain **one** way in which mitochondria are adapted to perform their function. [2]

.....

.....

.....



2. Meselson and Stahl investigated the replication of DNA prior to cell division.

In their experiments, bacteria were cultured in a medium containing only the ^{15}N isotope (heavy nitrogen). The bacteria were then transferred to ^{14}N (light nitrogen) culture medium and allowed to divide. Following each generation, samples of DNA were taken, separated using an ultracentrifuge and analysed.

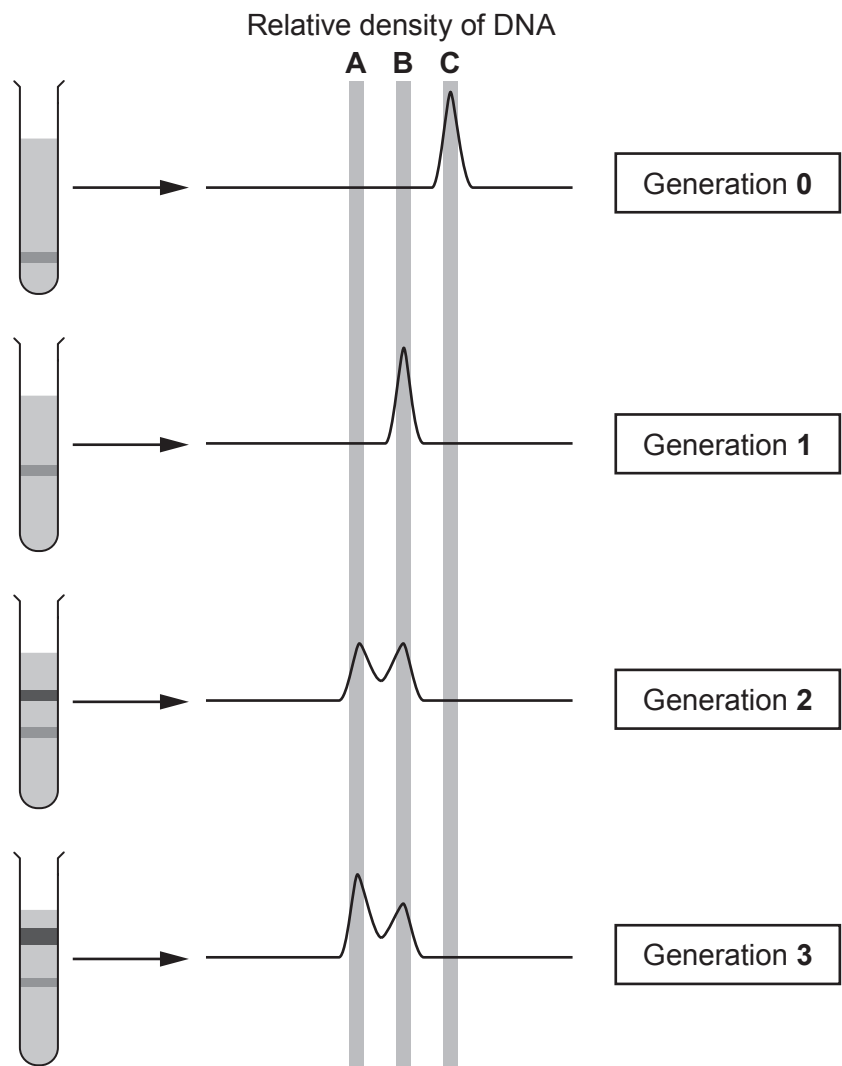
- (a) Identify **three** variables which would need to be controlled during centrifugation of the samples. [2]

.....

.....

.....

Photographs were taken of the extracted bands of DNA. The relative density of each band was scanned and measured; the more dense the DNA, the higher the reading. Shown below are some of their experimental results.



(b) (i) State which isotopes of nitrogen would be found in the DNA at **A**, **B** and **C** in the scans. [1]

A

B

C

(ii) Explain how the results in generations **0** and **1** support the theory of semi-conservative replication. [3]

.....

.....

.....

.....

.....

.....

.....

(iii) Explain the height of the peaks in the scans for generations **2** and **3**. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

2400U101
05



(iv) Name **two** enzymes now known to be involved in DNA replication and explain their functions. [3]

.....

.....

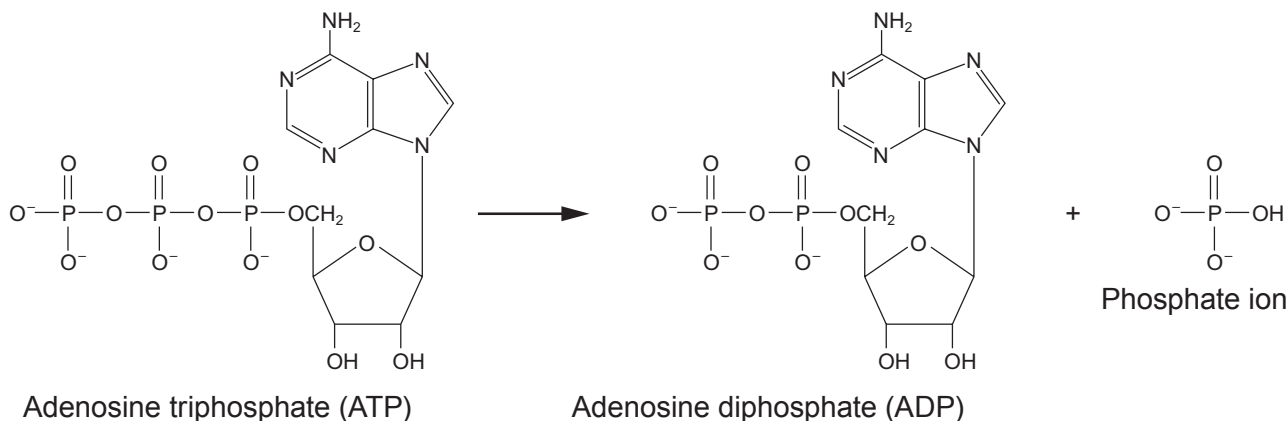
.....

Examiner
only

13



3. Adenosine triphosphate (ATP) belongs to a group of molecules that are made from nucleotides. ATP is an unstable molecule which possesses two high energy phosphate bonds and is often referred to as the universal energy currency. The diagram below shows the conversion of ATP into ADP.



- (a) (i) State the parts of the molecule that are common to DNA, RNA and ATP nucleotides. [1]

.....

.....

- (ii) State what is meant by the term *universal energy currency*. [1]

.....

.....

.....

- (iii) State whether the reaction shown above is exergonic or endergonic. Explain your answer. [2]

.....

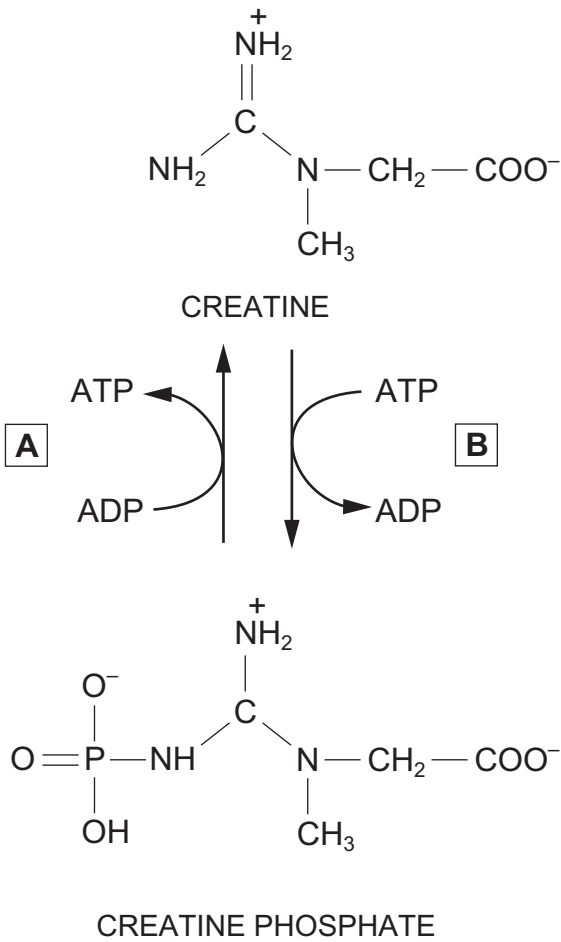
.....

- (b) It is estimated that the mass of ATP that is found within a resting human at any one time is approximately 5g, yet a human uses an estimated 50 kg of ATP each day. Calculate the percentage of the total ATP used per day that is available at any one time. [2]

Percentage of the total ATP available = %



Creatine phosphate is a naturally occurring chemical that works in partnership with ATP to increase the ability of muscle tissue to perform. It is not an energy source itself, but is used to increase the availability of ATP. It is stored inside muscle tissue.



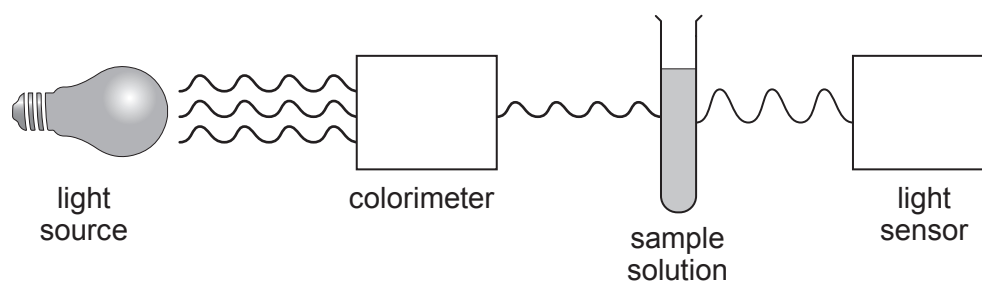
4. Lipases are involved in diverse biological processes, including the hydrolysis of triglycerides. Some lipases act in specific compartments inside cells, while others work in extracellular spaces.

(a) State the term given to enzymes which work within specific compartments inside cells.

[1]

.....

Triacetin is a colourless triglyceride and can be hydrolysed by bacterial lipase. When an indicator is added to alkaline triacetin solution a coloured solution is produced. A colorimeter can then be used to monitor the progress of the reaction. When the triacetin is completely digested, the solution will change from coloured to colourless, allowing more light to pass through, and produce an absorbance reading of zero.



The effect of temperature on the hydrolysis of triacetin was investigated.

The experiment was performed as follows:

1. 2 cm^3 of triacetin was mixed with 3 cm^3 of sodium hydrogen carbonate (pH10) in a test tube and 5 drops of indicator added.
2. 1 cm^3 of lipase was placed in a separate test tube.
3. Both test tubes were placed in a water bath (20°C) for 5 minutes.
4. The contents of the tubes were combined, mixed thoroughly and 2 cm^3 transferred to a cuvette.
5. The time taken for the absorbance to reach zero was measured.
6. The experiment was repeated to give a total of three readings.
7. Steps 1-6 were repeated at 25 , 35 , 45 , and 50°C .

(b) (i) Identify the independent and dependent variables in this investigation.

[2]

independent

dependent



(ii) Explain why the solutions were left in the water bath for 5 minutes before combining and mixing. [1]

.....

.....

.....

(iii) Explain why it would be inappropriate to use a buffer solution in this experiment. [3]

.....

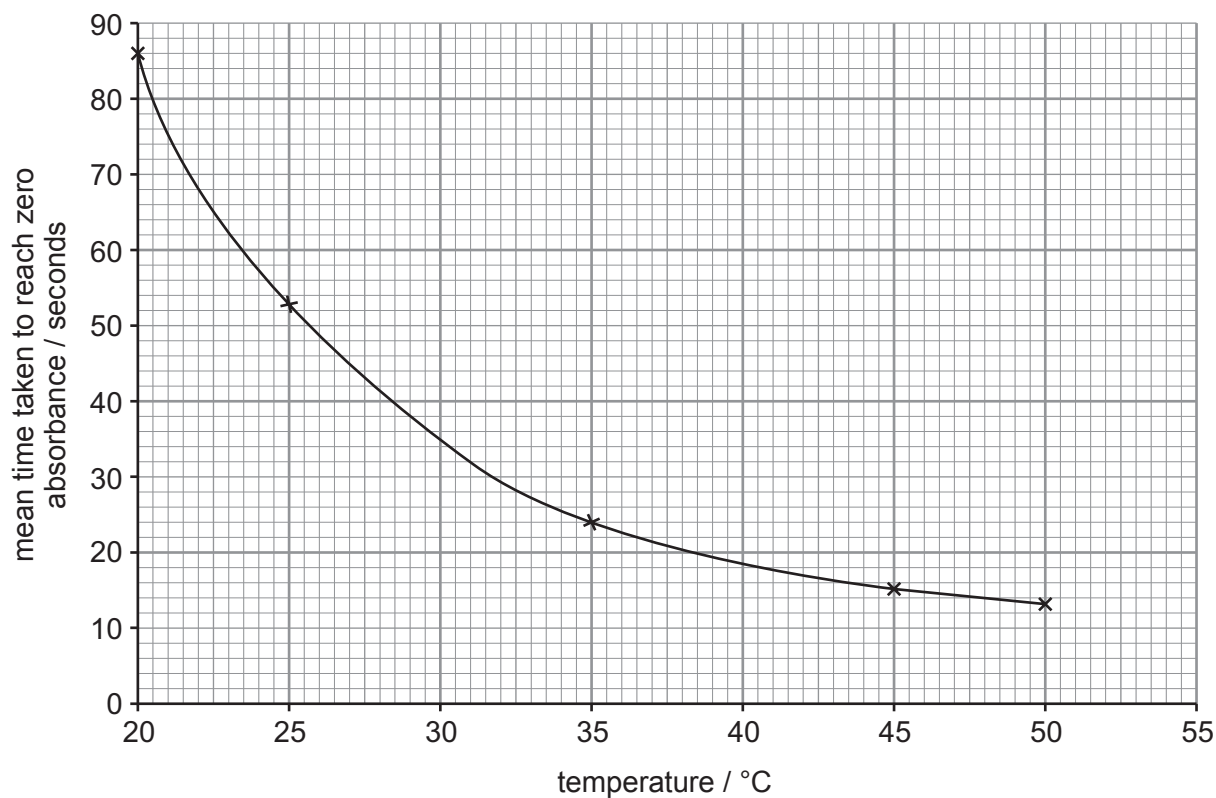
.....

.....

.....

.....

The results of the investigation are shown below.



Examiner
only

- (c) (i) In this experiment, 2 cm^3 of 0.25 g cm^{-3} triacetin was used. Calculate the rate of digestion per second at 40°C .

Give your answer in standard form. [3]

Rate = g s^{-1}

- (ii) Suggest **one** source of inaccuracy in this experimental method. [1]

.....

.....

.....

.....

- (iii) It was not possible to determine the optimum temperature for this enzyme in this experiment. Explain why and suggest how the design might be improved. [4]

.....

.....

.....

.....

.....

.....

.....

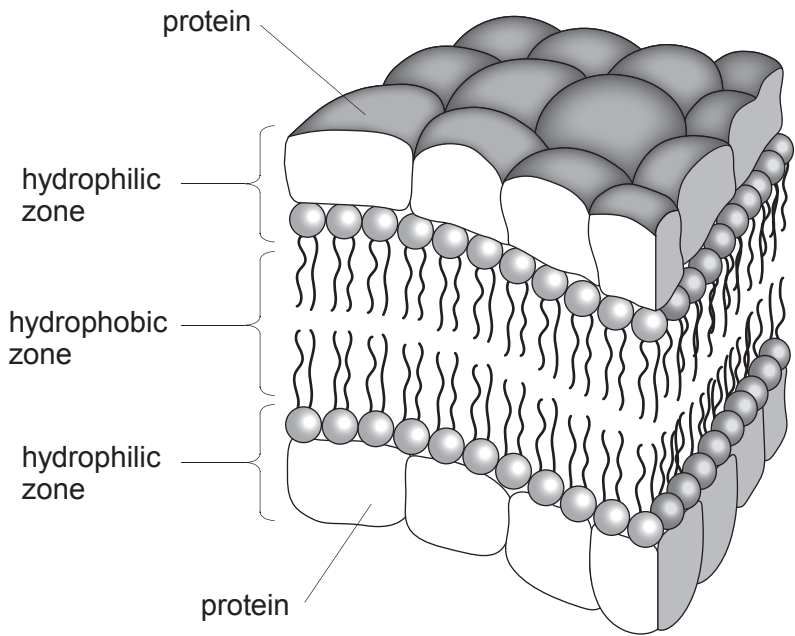
.....

.....

.....



5. In 1935, Hugh Davson and James Danielli proposed a model of the arrangement of molecules in the plasma membrane of a cell.



- (a) Compare Davson-Danielli's model shown in the diagram to the currently accepted fluid mosaic model of the plasma membrane. [4]

.....

.....

.....

.....

.....

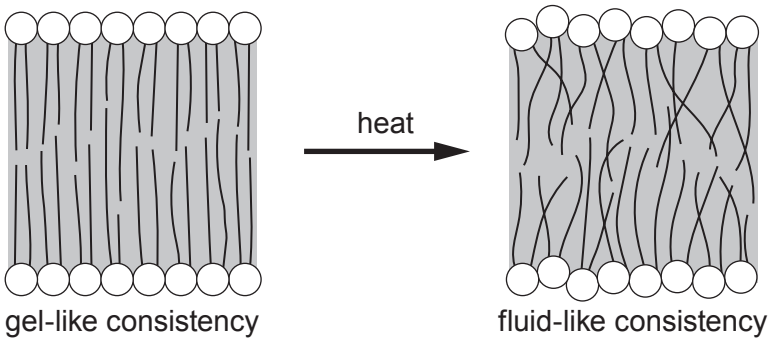
.....

.....

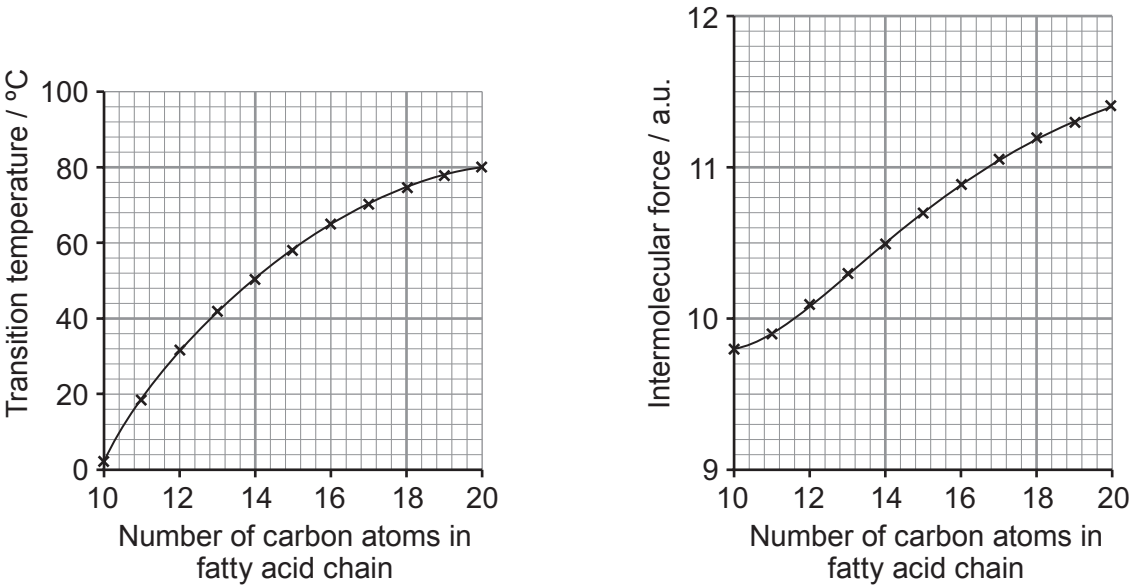
.....



Fluidity of the plasma membrane is due to the lateral movement of phospholipids. Increased fluidity occurs when the phospholipid bilayer changes from a more fixed gel-like consistency to a fluid-like consistency. Heating the molecules can result in increased fluidity.



The transition temperature, is the temperature at which the phospholipid bilayer changes from a gel like consistency to a fluid like consistency. This can also be affected by the length of the fatty acid chains which affect the strength of the intermolecular forces. These relationships are shown in the graphs below.



Examiner
only

- (b) (i) With reference to the graphs, describe the relationship between fatty acid chain length and the temperature at which the phospholipid bilayer becomes fluid. Explain your answer. [3]

.....

.....

.....

.....

.....

.....

.....

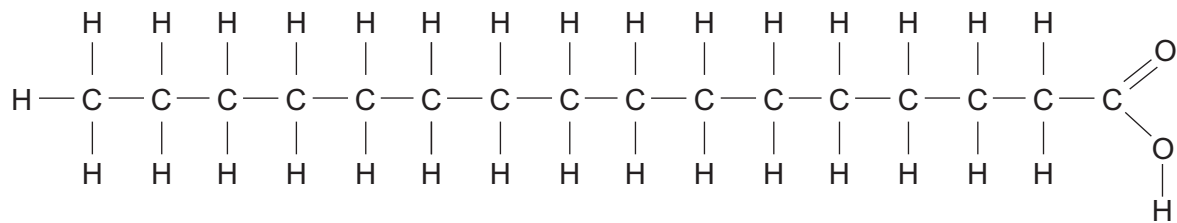
.....

.....

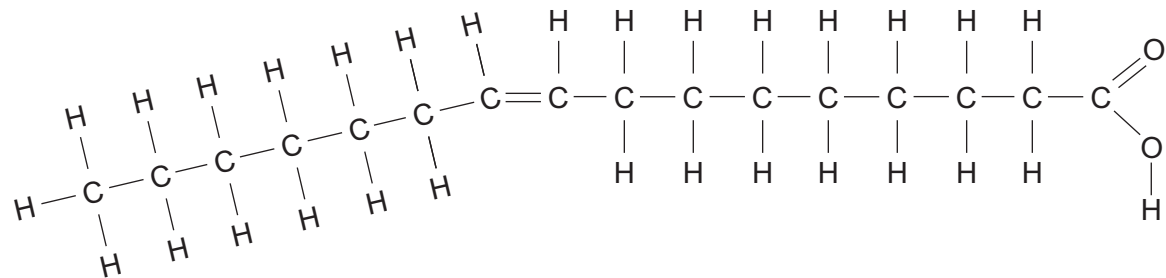


The packing of hydrocarbons also plays a significant role in affecting fluidity.
Two examples of fatty acids that can be found in phospholipids are shown below.

Saturated Fatty Acid



Unsaturated Fatty Acid



(ii) Suggest why the presence of unsaturated fatty acids in the phospholipid would increase fluidity of the membrane. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



Examiner
only

Cholesterol is a non-polar substance that is produced in the body and obtained from foods that come from animals (particularly egg yolks, meat, poultry, fish, and dairy products). The body needs this substance to build cell membranes. However, too much cholesterol increases a person’s risk of developing heart disease.

(iii) Suggest the position of cholesterol molecules within the plasma membrane. [1]

.....

.....

(iv) State how a diet high in animal products and cholesterol can lead to a person developing heart disease. [2]

.....

.....

.....

.....

.....

14



6. The cell surface membranes of root cells are responsible for differences in the concentration of ions in the roots when compared to those found in the surrounding soil solution. Inorganic ions are essential for cell metabolism in plants.

(a) Describe the role of magnesium, calcium, and phosphate ions in plant cells and tissues. [3]

.....

.....

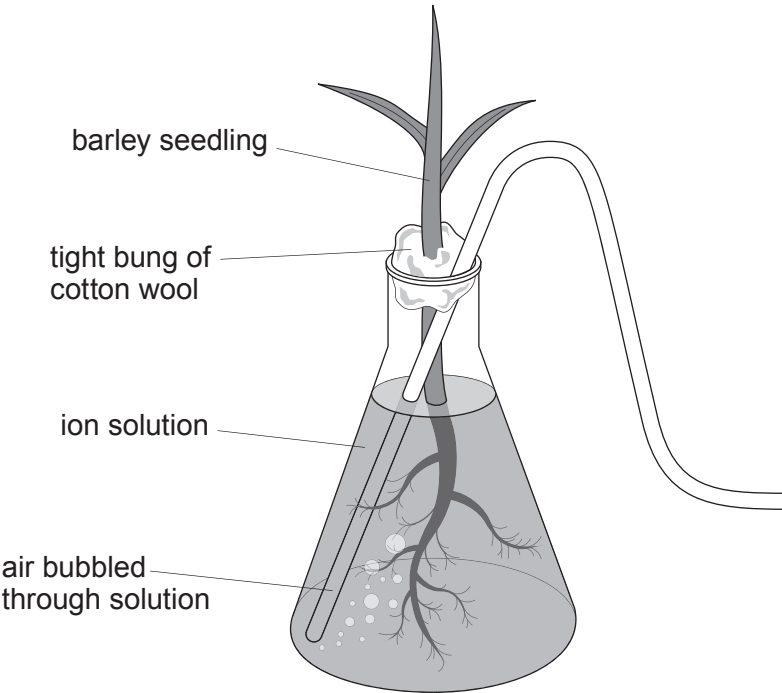
.....

.....

.....

.....

To investigate the uptake of mineral ions by plants, barley seedlings were grown with their roots in a solution of mineral ions for four days. Air was bubbled through the solution. The concentration of the ions in the solution and the root tissue were determined.



ion	ion concentration / mmol dm ⁻³	
	surrounding solution	root tissue
calcium (Ca ²⁺)	1.0	57.0
magnesium (Mg ²⁺)	0.3	33.0
phosphate (PO ₄ ³⁻)	0.9	1.6



Examiner
only

- (b) (i) With reference to the data, explain the relative concentrations in the surrounding solution and the root tissue after four days. [3]

.....

.....

.....

.....

.....

.....

- (ii) Explain why the growth of a barley seedling may be reduced if the solution did not have air bubbled through it. [4]

.....

.....

.....

.....

.....

.....

.....

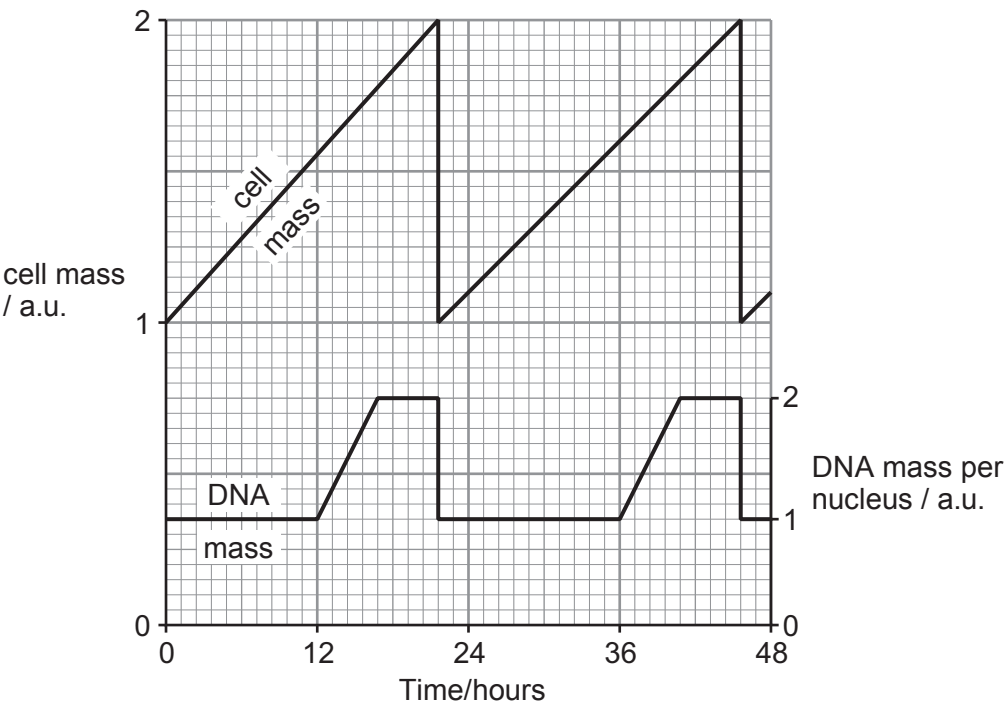
.....

10



Examiner
only

7. Cells can be grown under laboratory conditions and used in the development of drugs to treat cancer. The graph below shows a model of the general changes in cell mass and changes in mass of DNA in two cell cycles.



Identify the type of cell division shown above and explain the changes in cell mass and mass of DNA per nucleus in one cell cycle. [9 QER]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



Examiner
only

Area for handwritten answers, consisting of a large rectangle with horizontal dotted lines.



Examiner
only



Examiner
only

9

END OF PAPER



Surname	Centre Number	Candidate Number
Other Names		2



GCE AS/A LEVEL

2400U10-1



BIOLOGY – AS unit 1
Basic Biochemistry and Cell Organisation

THURSDAY, 24 MAY 2018 – AFTERNOON
1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	10	
2.	6	
3.	11	
4.	13	
5.	15	
6.	16	
7.	9	
Total	80	

ADDITIONAL MATERIALS

A calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the continuation pages at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

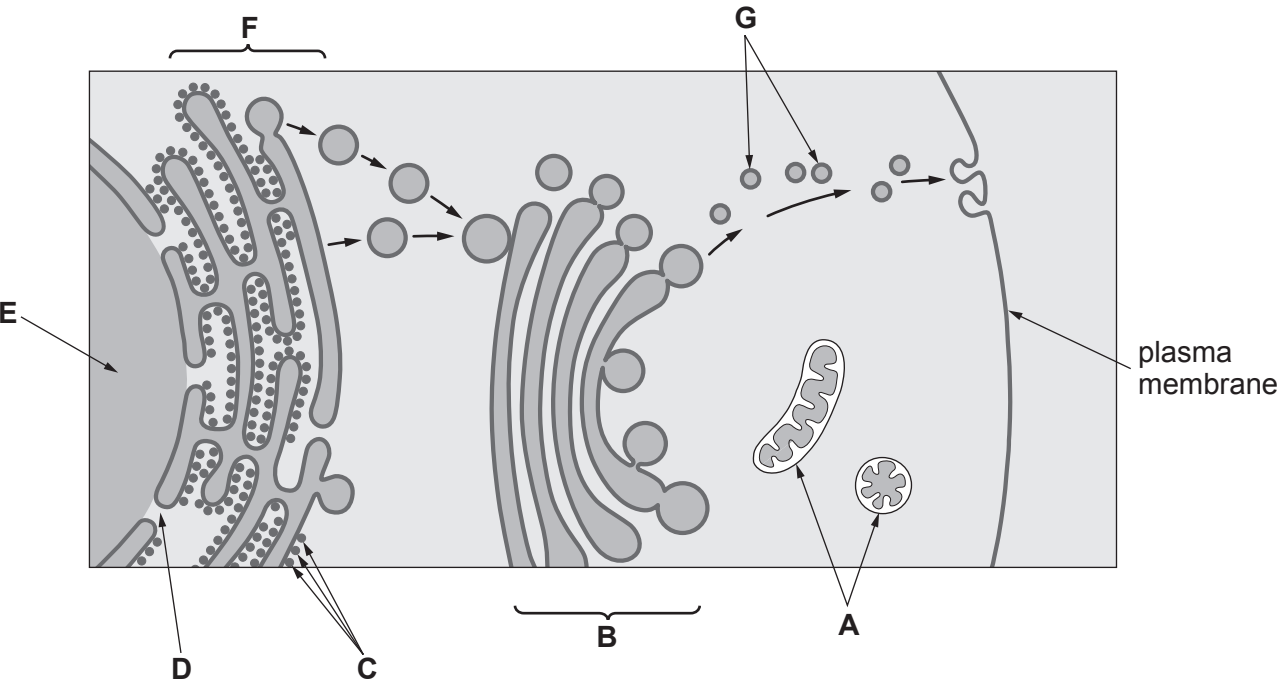
The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question 7.
The quality of written communication will affect the awarding of marks.



MAY182400U10101

Answer all questions.

1. The diagram below shows part of a cell taken from the pancreas, which is involved in the production of digestive enzymes.



(a) Identify the structures labelled **A**, **B**, **C**, **D**.

[2]

- A:**
B:
C:
D:



Examiner
only

(b) (i) With reference to the structures labelled **C, D and E**, describe the sequence of events that lead to the production of digestive enzymes in this cell. [3]

.....

.....

.....

.....

.....

.....

.....

(ii) With reference to the structures **B, F and G**, describe the sequence of events that lead to the secretion of digestive enzymes from this cell. [3]

.....

.....

.....

.....

.....

.....

.....

(iii) Explain the role of organelle **A** in the production and secretion of digestive enzymes. [2]

.....

.....

.....

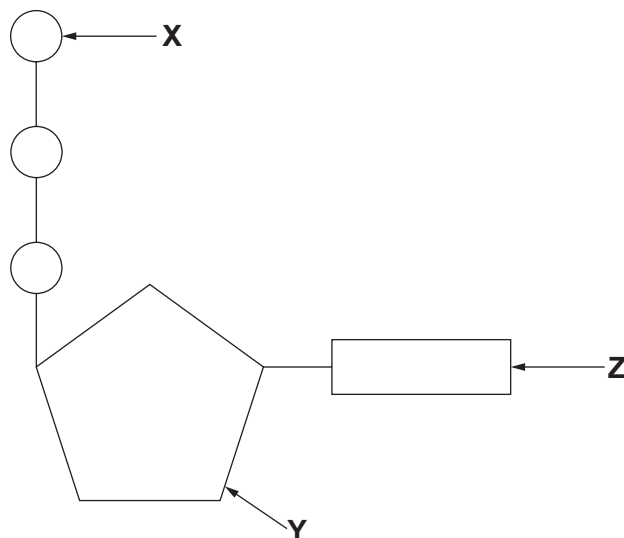
.....

10

2400U101
03



2. The diagram below shows a molecule of ATP which is produced during respiration and is an energy carrier molecule used in cells. When ATP is broken down into ADP and P_i , 30.6 kJ mol^{-1} of energy is released, which can be used for cellular activities.



- (a) (i) State the names of the parts of the ATP molecule labelled **X**, **Y** and **Z**. [2]

X:

Y:

Z:

- (ii) State **two** uses for the energy released from ATP in a plant cell. [1]

.....

.....

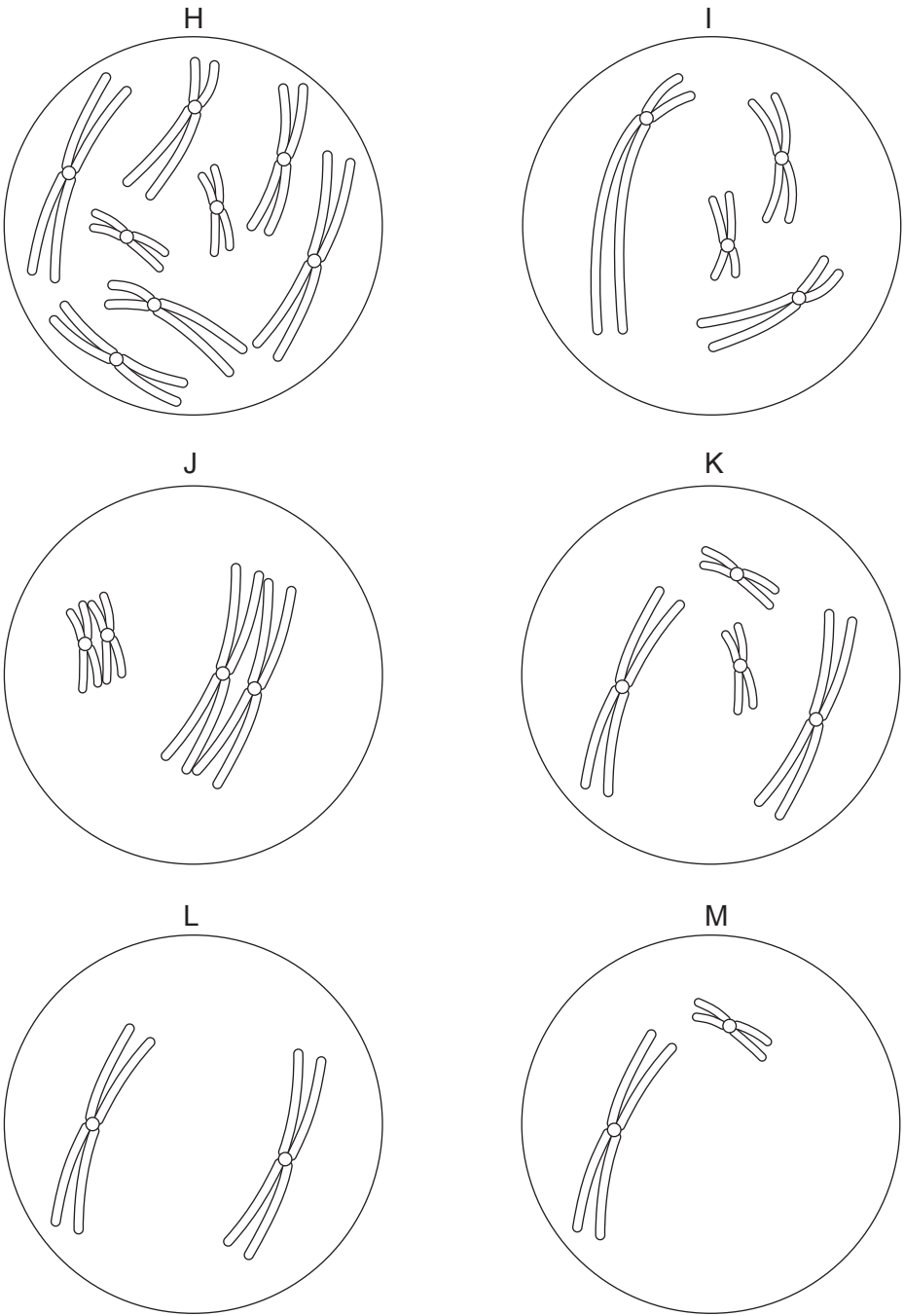
- (b) One mole of glucose releases 2870 kJ of energy when completely oxidised during aerobic respiration. Synthesis of one mole of ATP from ADP and P_i requires 30.6 kJ of energy. Calculate the percentage (%) efficiency of respiration if 38 moles of ATP are produced from one mole of glucose. Give your answer to three significant figures. [3]

% efficiency =



3. Spider mites are small invertebrates which are related to spiders and scorpions. One species, *Eutetranychus africanus*, which is native to the island of Mauritius, has very few chromosomes, $2n = 4$.

The diagrams below show the chromosomes in some cells undergoing mitosis or meiosis.

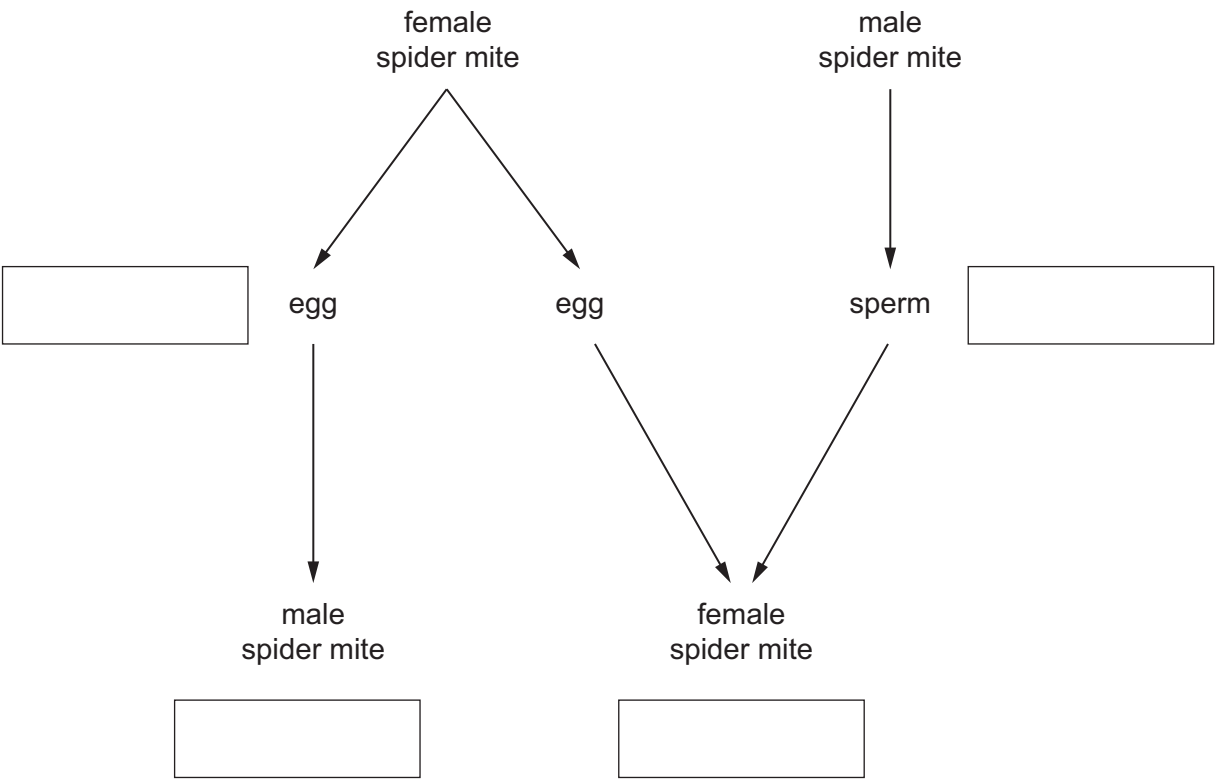


(a) Identify the cells that belong to this species of spider mite.

[2]



(b) The diagram below shows the individuals that make up a colony of spider mites. Female spider mites are produced when a male sperm fertilises a female's egg. Male spider mites develop from an unfertilised female egg.



- (i) **Complete** the diagram to state which stages are **haploid** or **diploid**. [2]
- (ii) What conclusions can be made about the formation of male and female gametes in spider mites? [2]

.....

.....

.....



- (c) (i) A sample of tissue from the **leg muscle** of a young spider mite was analysed. The mass of DNA in some of the cells was 6.8 arbitrary units (au), whilst in other cells it was 3.4 au. Suggest explanations for this difference using your knowledge of the cell cycle. [3]

.....

.....

.....

.....

- (ii) The table below shows the percentage of cells, with either 3.4 au or 6.8 au of DNA, in the leg muscle of a young spider mite and an older spider mite.

	Percentage of cells	
Mass of DNA in cell /au	Young spider mite	Older spider mite
6.8	20	5
3.4	80	95

What conclusions can be drawn from the data regarding the significance of mitosis in these spider mites? [2]

.....

.....

.....

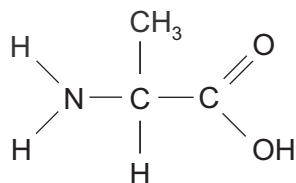
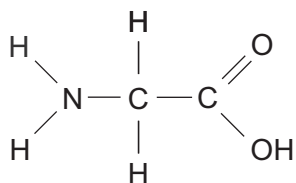
.....



4. β -Bungarotoxin is a neurotoxin found in the venom of Krait snakes. The neurotoxin is a protein that causes muscle paralysis and eventual death.

The diagram below shows the structure of two amino acids found within the protein.

- (a) (i) **Complete** the diagram, to show the products formed when these amino acid molecules are joined by a condensation reaction. [2]



- (ii) State the bond formed.

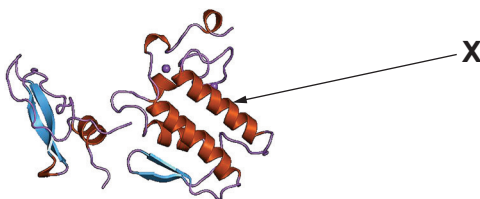
[1]

.....



Examiner
only

- (b) The diagram below shows the structure of the β -bungarotoxin. The protein is composed of two polypeptide chains.



- (i) State the highest level of protein structure exhibited by the β -bungarotoxin. [1]

- (ii) Name the structure labelled **X** and state how this structure is maintained. [2]

- (iii) State the minimum number of genes required to code for the β -bungarotoxin. Explain how you reached this conclusion. [1]

- (c) (i) One polypeptide chain of β -bungarotoxin contains 120 amino acids. State the minimum number of DNA nucleotides that would be required to code for this polypeptide. [1]

- (ii) Scientists have isolated, and sequenced, the gene for this polypeptide chain; it was found to contain over 2000 nucleotides. Scientists have also isolated molecules of mRNA, which code for this polypeptide, and found that some mRNA molecules had a higher molecular mass than the others. What conclusions can be drawn about the production of this polypeptide? [5]



5. *Nereis virens* is a species of marine worm. A student wanted to investigate the effect of a change in solute concentration on these worms. The following method was used.
- Collect seawater and thirty worms.
 - Make a 10% solution of seawater by placing 1 dm³ of seawater and 9 dm³ of freshwater into a fish tank.
 - Record the mass of ten worms and place them into the fish tank of diluted seawater.
 - Every 15 minutes remove the worms, blot dry, re-weigh and record their mass.
 - Repeat this process, and after 75 minutes, return the worms to a container of undiluted seawater.
 - Repeat the experiment a further two times using different worms for each trial.
 - Calculate the mean percentage change in mass of the worms during the investigation.

The student’s results are shown in the table below.

Time / minutes	Mean percentage change in mass / %
0	0
15	15
30	18
45	20
60	22
75	22

(a) Explain why it was necessary for the student to calculate the percentage change in mass. [2]

.....

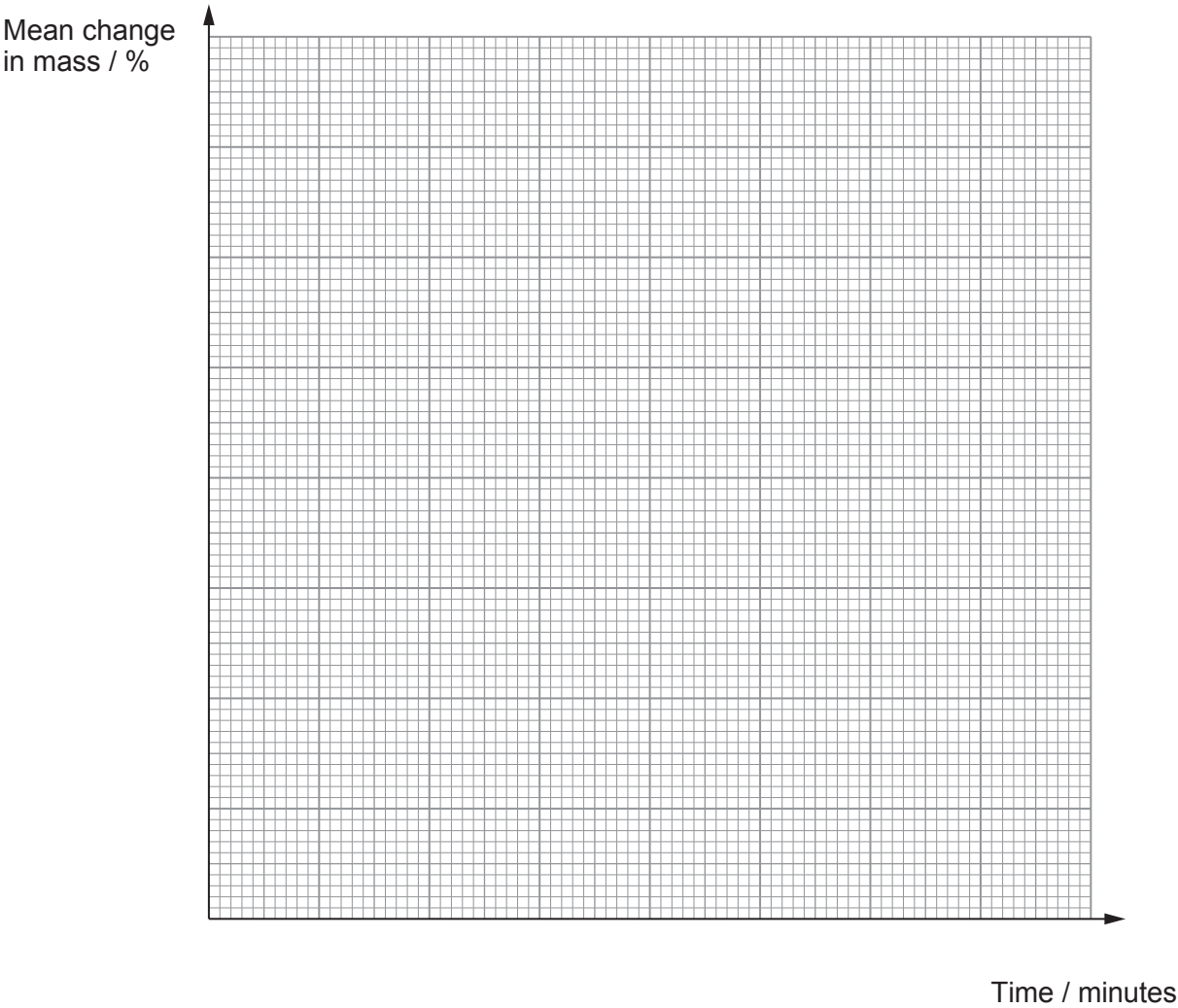
.....

.....



(b) (i) Plot the student's results shown in the table on the grid below.

[3]



(ii) Describe and explain the changes in mass of the worms during the experiment. [4]

.....

.....

.....

.....

.....

.....

.....

.....



Examiner
only

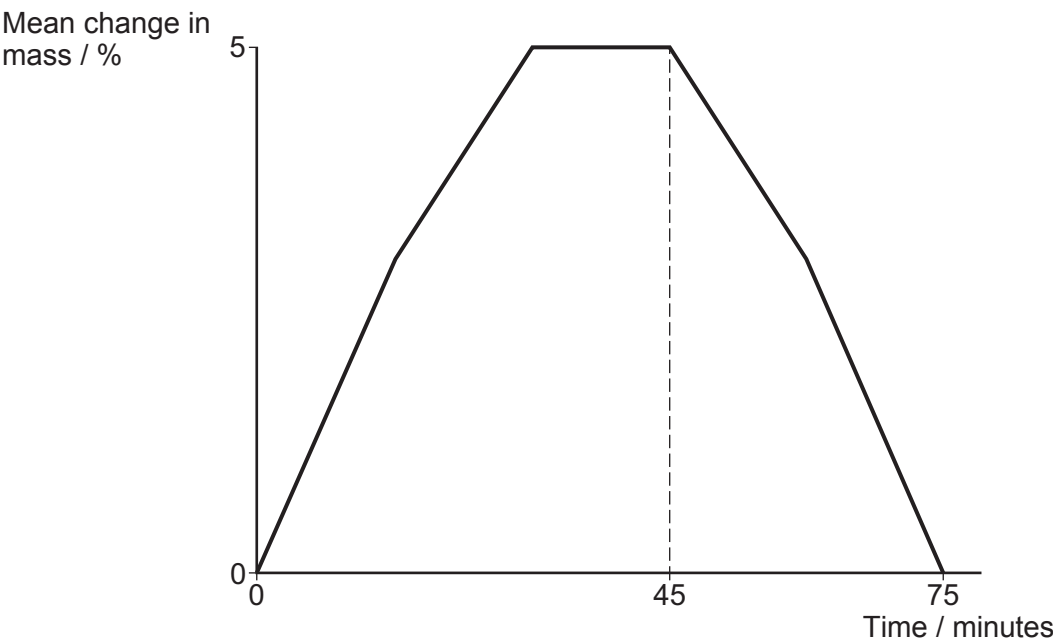
- (c) The student decided to repeat the experiment. She collected ten fresh worms, weighed them and placed them in a fish tank containing undiluted seawater. Explain why it was important for the student to carry out this second experiment. [2]

.....

.....

.....

- (d) A similar experiment was set up with a different species of marine worm, *Golfingia gouldi*. The results are shown below.



- (i) Describe how the results for *Golfingia* differ from those of *Nereis*. [2]

.....

.....

.....

.....



(ii) *Golfingia* can actively pump ions from its cells into the surrounding water. Explain the change in mass of *Golfingia* between 45 and 75 minutes. [2]

.....

.....

.....

.....

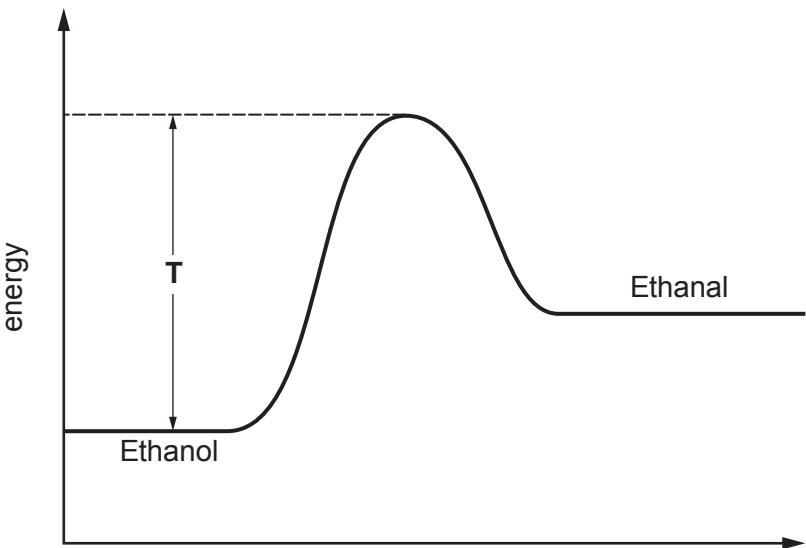
Examiner
only

15



6. Alcohol dehydrogenase is an enzyme that catalyses the conversion of ethanol into ethanal.

The graph below shows the change in energy state when ethanol is converted to ethanal when no enzyme is present.



(a) (i) State what is represented by the region labelled **T**. [1]

(ii) Draw a curve on the diagram to show the change in energy level when alcohol dehydrogenase is present. [1]

(b) Describe how alcohol dehydrogenase can catalyse the conversion of ethanol into ethanal. [3]

.....

.....

.....

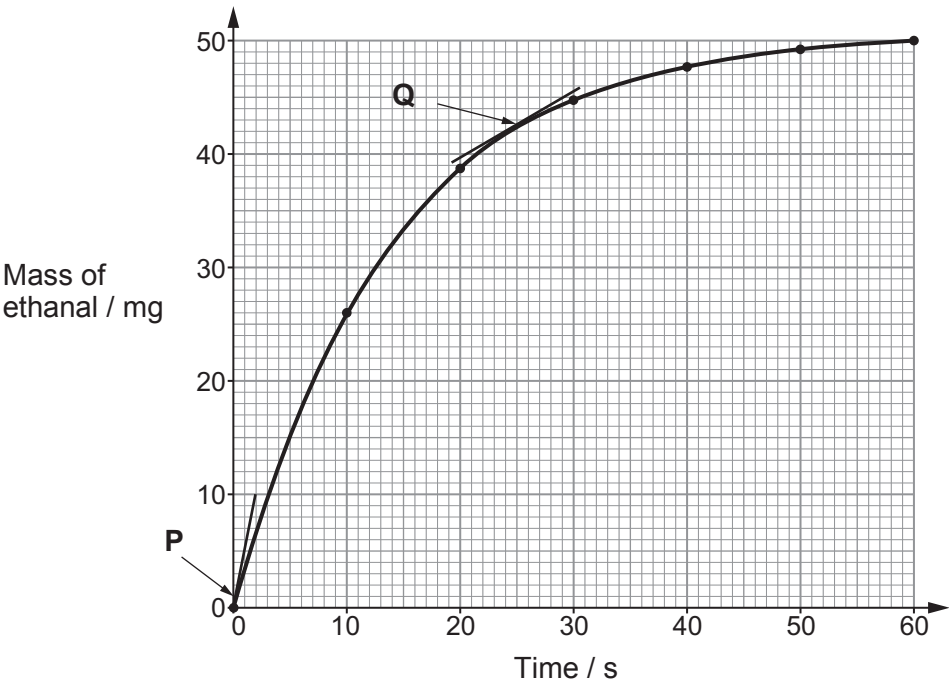
.....

.....

.....



- (c) A fixed mass of ethanol was added to a test tube containing alcohol dehydrogenase and a pH7 buffer solution. The test tube was incubated at 30°C and the mass of ethanal produced over time was recorded. The results are shown below.



- (i) The rate of reaction at **P** was 5 mg s^{-1} .
Calculate the rate of reaction at **Q**. [2]

rate = mg s^{-1}

- (ii) What conclusions can be drawn that would account for the difference in rate at **P** and at **Q**? [4]

.....

.....

.....

.....

.....

.....

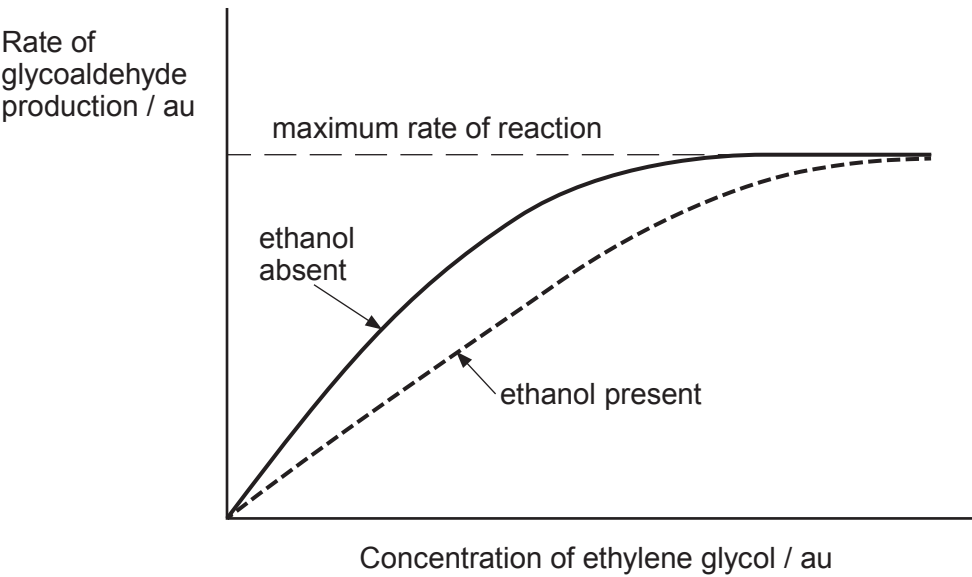


Examiner
only

(d) Ethylene glycol is a colourless, odourless, sweet liquid, commonly found in antifreeze. It is highly toxic if ingested because once inside the body ethylene glycol is converted into glycoaldehyde. This reaction is also catalysed by alcohol dehydrogenase.

Treatment of ethylene glycol poisoning includes giving the patient ethanol, either intravenously or orally.

The graph below shows the rate of glycoaldehyde production in the presence, and absence of ethanol.



Use the information provided to explain why this treatment would reduce the effects of ethylene glycol poisoning. [5]

.....

.....

.....

.....

.....

.....

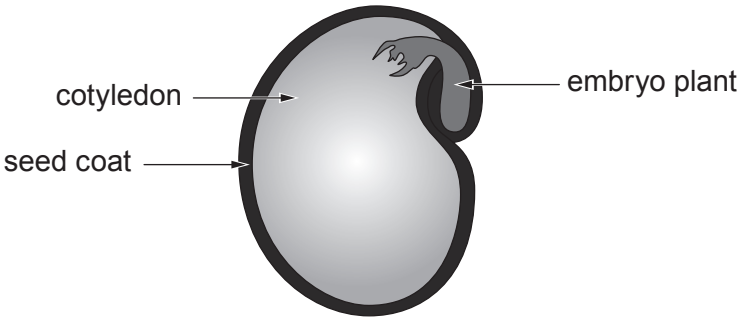
.....

.....



Examiner
only

7. The diagram below shows the generalised structure of a seed.



The cotyledon contains many organic molecules including starch and triglycerides. Describe the structure of starch and triglycerides and suggest how their structure and properties relate to their function as energy storage molecules. [9 QER]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



Examiner
only

Area for student response with horizontal ruling lines.



Examiner
only

9

END OF PAPER



Surname	Centre Number	Candidate Number
Other Names		2



GCE AS/A LEVEL

2400U10-1



BIOLOGY – AS unit 1 Basic Biochemistry and Cell Organisation

TUESDAY, 21 MAY 2019 – AFTERNOON

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	10	
2.	15	
3.	14	
4.	18	
5.	14	
6.	9	
Total	80	

2400U101
01

ADDITIONAL MATERIALS

A calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional pages at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question 6.

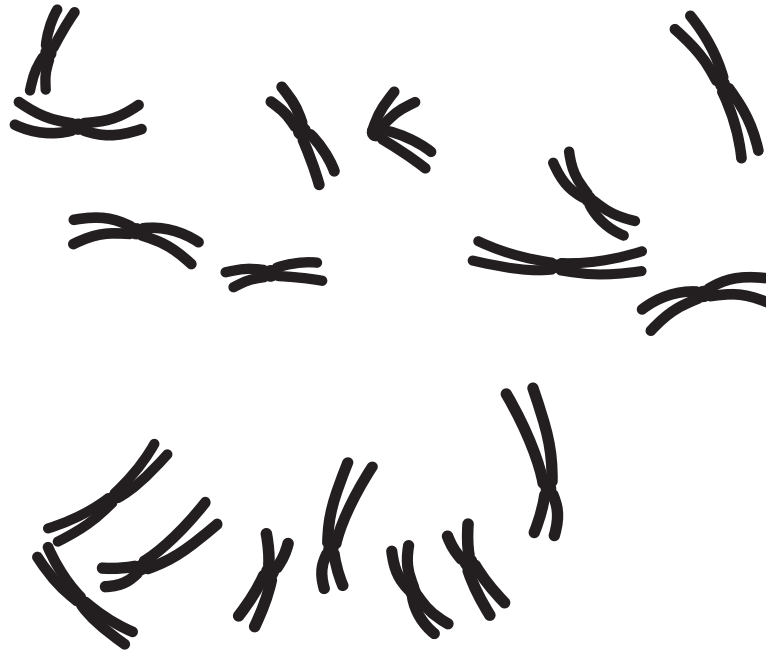
The quality of written communication will affect the awarding of marks.



MAY192400U10101

Answer all questions.

1. (a) The diagram below shows the chromosomes from a single cell of a Chinese daisy, *Aster shennongjiaensis*, during prophase of mitosis.



- (i) State the **haploid** number of chromosomes in this species. [1]

.....

- (ii) State the evidence from the diagram that the cell is undergoing mitosis and not meiosis. [1]

.....

.....

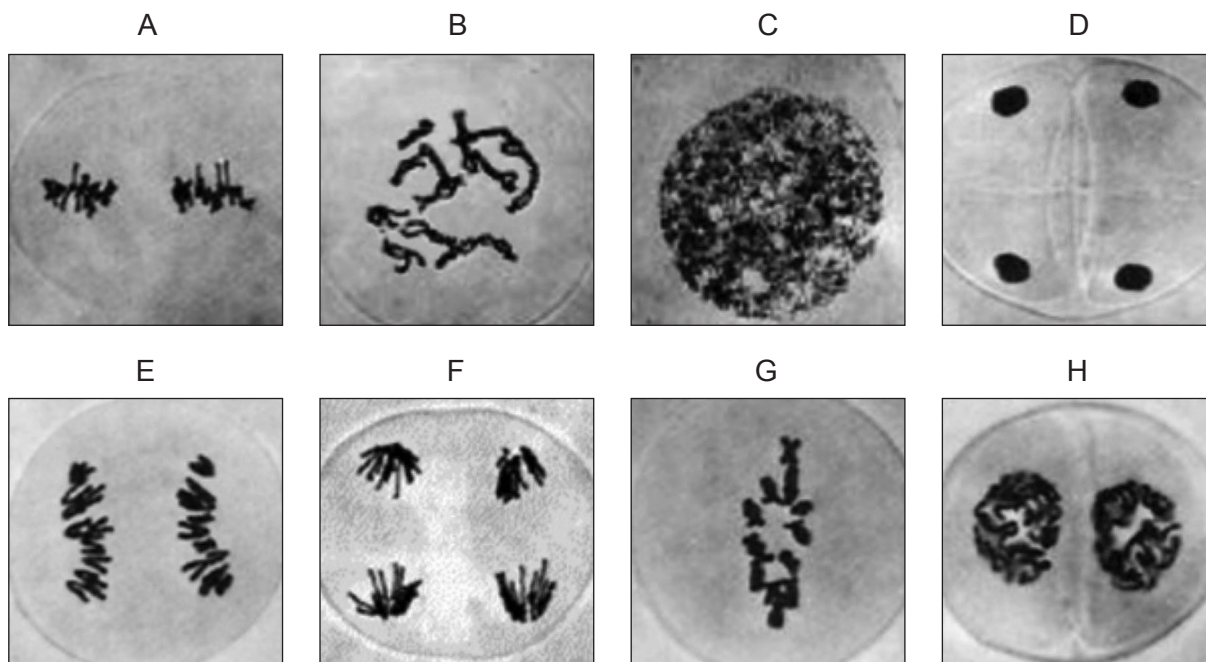
- (iii) When the tissues from the plant were removed they were stained before being viewed under the microscope. Explain how this makes the chromosomes visible. [1]

.....

.....



(b) The photomicrographs below show different stages of meiosis in a different plant species.



- (i) State which organ of the plant could have been used to obtain the photomicrographs above. [1]

- (ii) Place the photomicrographs in the correct order. Some letters have been done for you. [1]

C					A	F	D
---	--	--	--	--	---	---	---

- (iii) Use your knowledge of cell division to describe **two** differences between the stages labelled **E** and **F**. [2]

.....

.....

.....

.....



- (iv) With reference to photomicrographs **B** and **G**, state how meiosis contributes to genetic variation in the species and explain why this is important. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

10

2400U101
05

-
- The diagram compares the internal structures of a plant cell (Chlorella) and an animal cell (Amoeba).
- Chlorella (Plant Cell):**
- A:** Points to the chloroplasts, which are oval-shaped organelles with internal stacks of thylakoids.
 - B:** Points to the nucleus, a large, dark, spherical structure containing a nucleolus.
 - cellulose cell wall:** The thick, outermost layer of the cell.
- Amoeba (Animal Cell):**
- nucleus:** A large, dark, spherical structure.
 - contractile vacuole:** A small, clear, oval-shaped organelle.
 - plasma membrane:** The thin, outermost layer of the cell.

- [1]

organelle **B**:

- (i) Use your knowledge of osmosis to explain why contractile vacuoles are necessary for the survival of the *Amoeba*. [4]



- (ii) Identify **two** *other* organelles which must be present in large numbers in the *Amoeba* in order for this process to occur. Explain your answer for each organelle. [4]

Organelle 1

Explanation

.....

Organelle 2

Explanation

.....

- (c) Although *Chlorella* lives in the same habitat as *Amoeba*, it does not possess a contractile vacuole. Describe and explain how the structure of the cell wall of *Chlorella* allows it to survive in this environment. [4]

.....

.....

.....

.....

.....

.....

.....

.....

- (d) Bacteria, such as *Cyanophyceae* also inhabit fresh water ponds. Describe **two** differences between the structure of *Chlorella* and *Cyanophyceae*. [2]

.....

.....

.....

.....

.....



3. (a) A student wanted to investigate the concentration of casein in milk powder. Casein is an insoluble protein found in milk. When milk powder is mixed with water it forms an opaque (cloudy) suspension. The enzyme trypsin breaks down the casein into soluble peptides, forming a transparent solution.

To begin with the student needed to construct a calibration curve from known concentrations of milk powder suspensions. He was provided with the following:

- 10 cm³ suspensions made from milk powder at concentrations of 1%, 2%, 3%, 4% and 5% in separate test tubes with a buffer
- 1% trypsin solution

The milk suspensions and trypsin solution were placed separately in a thermostatically controlled water bath at 30 °C. Once both the milk suspensions and trypsin solution were at the correct temperature, he transferred 2 cm³ of trypsin to each of the suspensions of milk powder and timed how long it took for the suspensions to become transparent.

- (i) Using your knowledge of enzymes, explain why:

I. it was necessary for a buffer to be added to the solutions;

[2]

.....

.....

.....

II. all test tubes were placed in a thermostatically controlled water bath.

[2]

.....

.....

.....

The table below shows the results.

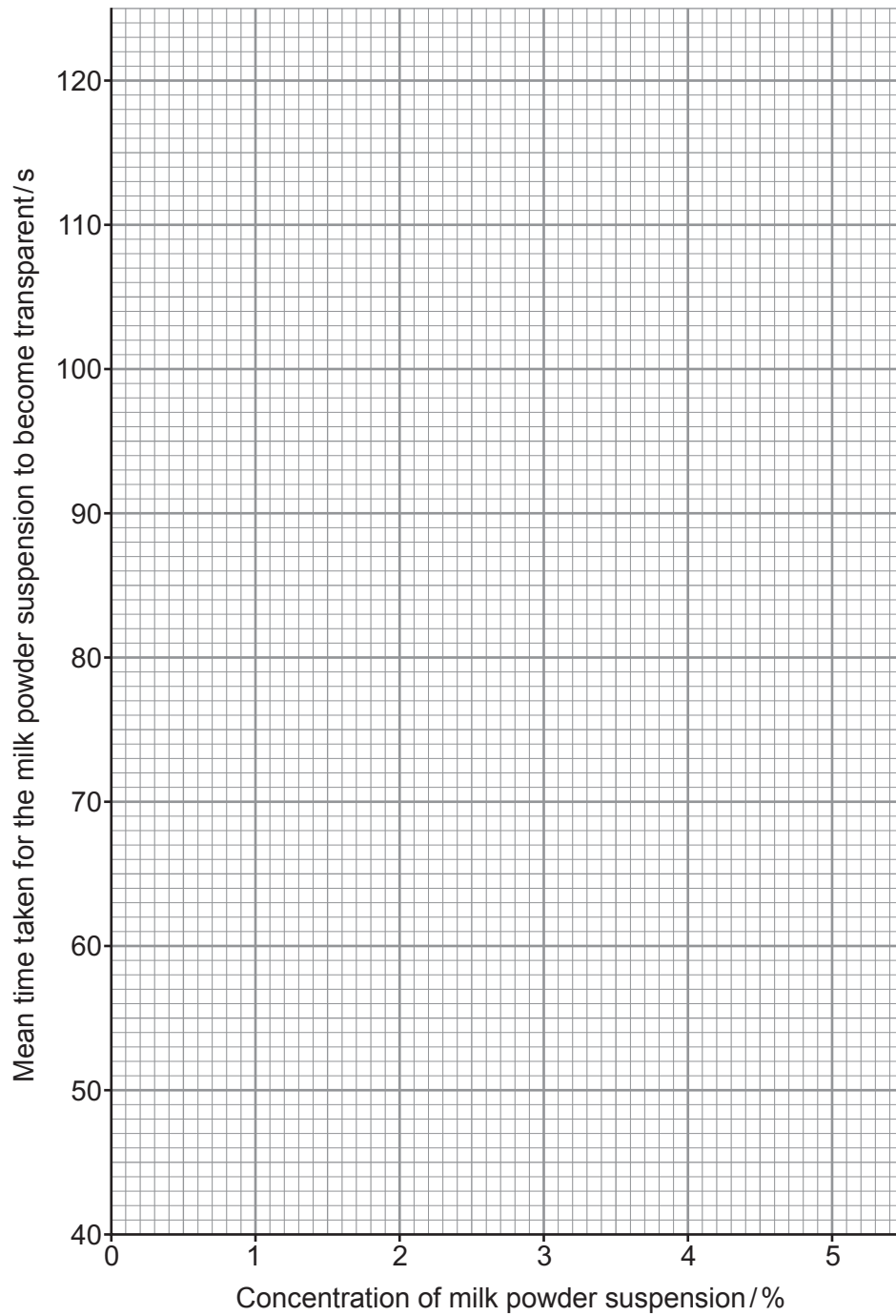
Concentration of milk powder suspension / %	Time taken for milk powder suspension to become transparent / s					
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	mean
1	49	51	55	52	53	52.0
2	66	64	62	61	63	63.2
3	88	77	73	81	84	80.6
4	91	87	93	86	99	
5	96	109	108	111	105	105.8

- (ii) Calculate the mean time taken for the 4% milk powder suspension to become transparent. **Write your answer in the table.**

[1]



- (iii) Plot the mean data on the graph paper provided. **Range bars should also be added to the graph.** [4]



- (b) The student was then provided with two different suspensions of milk powder, **X** and **Y**, of unknown casein concentration. He carried out the same procedure and obtained the following mean results: **X** took 61 seconds to clear and **Y** took 97 seconds to clear.

- (i) Estimate the percentage concentration of casein in these two suspensions. [1]

X: **Y:**

- (ii) The student had more confidence in the estimate for suspension **X** than the estimate for suspension **Y**. Use the graph to explain the reason for this statement. [2]

.....

.....

.....

.....

- (iii) Suggest **two** sources of inaccuracy in this experiment. [2]

.....

.....

.....

.....



4. Haemoglobin exhibits quaternary structure and consists of four subunits, two subunits called alpha-globin and two subunits called beta-globin.

(a) Define quaternary structure.

[1]

.....

.....

- (b) (i) The beta-globin molecule consists of 146 amino acids. State the minimum number of mRNA nucleotides required to code for this molecule.

[1]

.....

- (ii) The genetic code is a triplet code and is degenerate. The table below shows the **mRNA codons**, that code for specific amino acids.

		Second letter					
		U	C	A	G		
First letter	U	UUU Phenyl-alanine UUC UUA Leucine UUG	UCU Serine UCC UCA UCG	UAU Tyrosine UAC UAA Stop codon UAG Stop codon	UGU Cysteine UGC UGA Stop codon UGG Tryptophan	Third letter	U C A G
	C	CUU Leucine CUC CUA CUG	CCU Proline CCC CCA CCG	CAU Histidine CAC CAA Glutamine CAG	CGU Arginine CGC CGA CGG		U C A G
	A	AUU Isoleucine AUC AUA AUG Methionine initiation codon	ACU Threonine ACC ACA ACG	AAU Asparagine AAC AAA Lysine AAG	AGU Serine AGC AGA Arginine AGG		U C A G
	G	GUU Valine GUC GUA GUG	GCU Alanine GCC GCA GCG	GAU Aspartic acid GAC GAA Glutamic acid GAG	GGU Glycine GGC GGA GGG		U C A G

Use the table to explain why it is necessary that the code is a triplet code and why the code is said to be degenerate.

[2]

.....

.....

.....

.....

.....

.....



- (iii) The table below shows part of the DNA sequence that codes for beta-globin and the corresponding mRNA codons, tRNA anticodons and the amino acid sequence of the molecule.

Complete the table.

[4]

	Codon position number					
	3	4	5	6	7	8
DNA coding strand	GAC			CTC	CTC	
mRNA codons		ACU		GAG	GAG	
tRNA anticodons			GGA	CUC	CUC	
Amino acid sequence				Glutamic acid	Glutamic acid	Lysine

- (c) Describe the events which take place in the production of the beta-globin molecule after the ribosome has attached to the mRNA. **[5]**

.....

.....

.....

.....

.....

.....

.....

.....

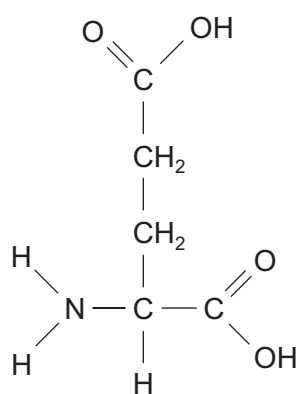
.....

.....

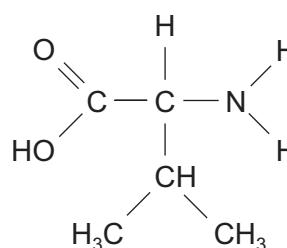


- (d) A mutation, where a single base was substituted by a different base, resulted in glutamic acid being replaced by valine at position 6 in the beta-globin molecule. Use the information provided to determine the mutated DNA base sequence at position 6 on the coding strand of DNA. [1]

- (e) The diagrams below show the structures of glutamic acid and valine.



glutamic acid



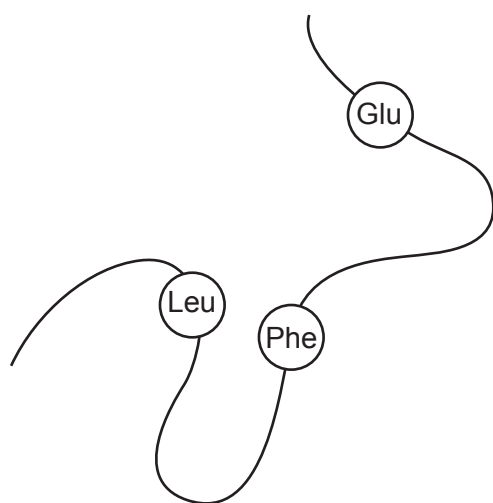
valine

- (i) **Draw a circle** around the R-groups of these **two** amino acids. [1]

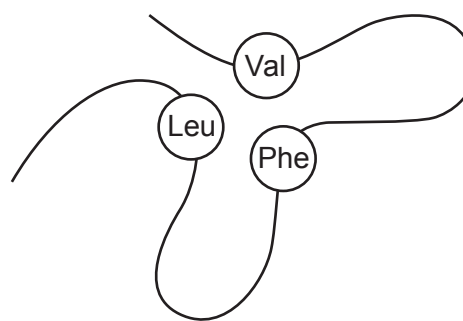
The R-group of glutamic acid is polar whereas the R-group of valine is non-polar. This affects the 3D structure of the resulting beta-globin molecule, as shown in the diagram below.

Key:

Glu – glutamic acid
Val – valine
Leu – leucine
Phe – phenylalanine



Section of beta-globin molecule in a person **without** mutation



Section of beta-globin molecule in a person **with** mutation



- (ii) Valine interacts with phenylalanine and leucine in the beta-globin molecule. Phenylalanine and leucine both have non-polar R-groups. Explain how the properties of these amino acids mean that valine interacts with phenylalanine and leucine but glutamic acid does not. [3]

.....

.....

.....

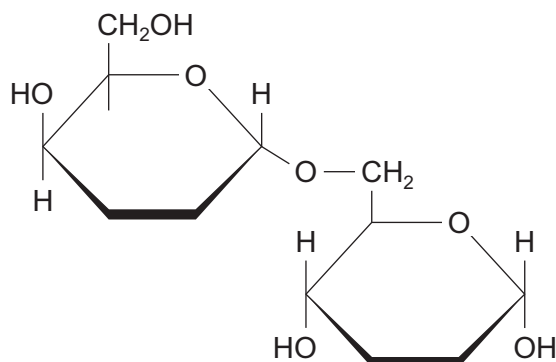
.....

.....



5. Melibiose is a reducing sugar that is formed from galactose and glucose and is thirty times sweeter than sucrose.

(a) The diagram below shows the structure of melibiose.



- (i) **Complete the diagram above** to show how this molecule would be broken down into two monosaccharides. [2]
- (ii) State the type of reaction that occurs when this molecule is broken down into two monosaccharides. [1]

.....



- (iii) Describe a biochemical test that could be used to determine the presence of a reducing sugar such as melibiose. [2]

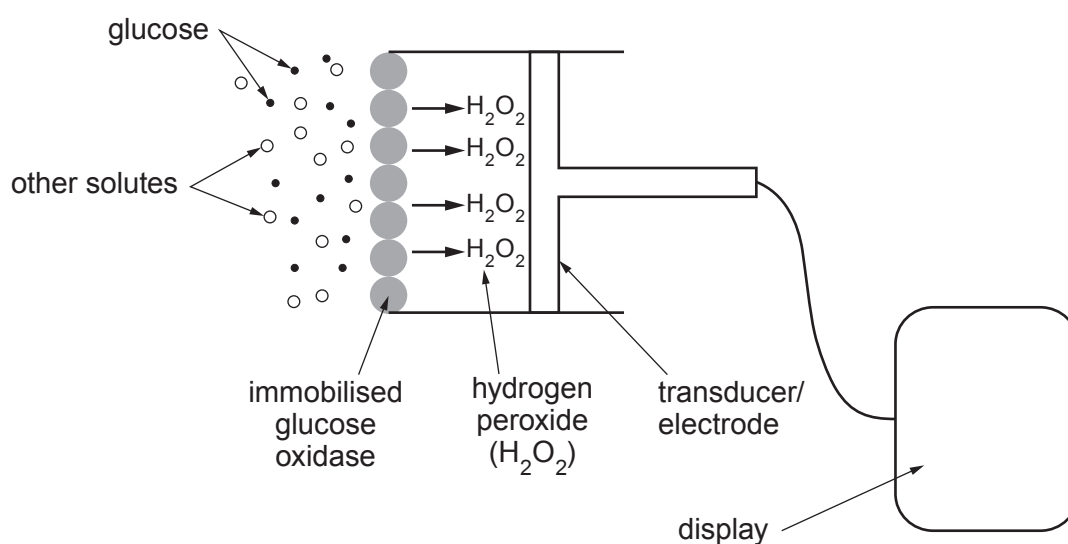
.....

.....

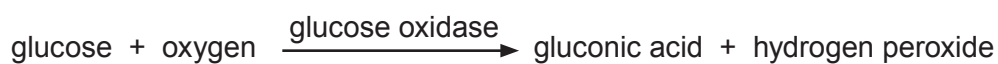
.....

.....

- (b) Biosensors make use of immobilised enzymes to detect specific molecules in a mixture. The diagram below shows a possible structure of a biosensor that could be used to detect the presence of glucose in a urine sample.



Within the biosensor, the glucose oxidase catalyses the following reaction:



The concentration of hydrogen peroxide is detected by the transducer and converted into an electrical signal, shown as a measurement on the display.



- (i) Using your knowledge of enzymes explain why this biosensor will only detect the presence of glucose. [3]

.....

.....

.....

.....

.....

.....

- (ii) With reference to the information given, explain how the biosensor could be used to give a measurement of the glucose concentration. [2]

.....

.....

.....

.....



- O[C@H]1O[C@H](O)[C@H](O)[C@@H](O)[C@H]1O

Experiments have shown that arabinose can inhibit glucose oxidase and therefore affect the accuracy of the biosensor. Using the information provided in part (a) and the diagram above, explain why the contamination of a urine sample with arabinose would affect the accuracy of the biosensor.

[4]



6. The following extracts are adapted from articles published on the Internet.

Hidden ocean found on Saturn's icy moon, Enceladus, could potentially support life.

Enceladus harbours a big ocean of liquid water beneath its icy crust that may be capable of supporting life as we know it. Researchers said that the water is about 10 kilometres deep and lies beneath a shell of ice 30 to 40 km thick. Furthermore, it is in direct contact with a rocky sea floor, theoretically making possible all kinds of complex chemical reactions – such as the kind that may have led to the rise of life on Earth.

NASA finds more evidence that the ocean on Enceladus could support alien life.

NASA's Cassini spacecraft orbiting the moon has analysed the plumes of gas forced out through fissures in the ice. These plumes have been found to contain four of the six most important elements of life on Earth – carbon, hydrogen, nitrogen and oxygen – only phosphorus and sulfur have not been detected.

Describe the structure of water and explain how the properties of water would be essential to supporting life on Enceladus. Explain how the absence of phosphorus and sulfur would prevent the formation of biochemical molecules essential for life on Earth. [9 QER]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



Handwriting practice area with 20 horizontal dotted lines.





END OF PAPER



Surname	Centre Number	Candidate Number
First name(s)		2



GCE AS/A LEVEL

2400U10-1



FRIDAY, 20 MAY 2022 – AFTERNOON

BIOLOGY – AS unit 1
Basic Biochemistry and Cell Organisation

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	10	
2.	7	
3.	11	
4.	14	
5.	14	
6.	15	
7.	9	
Total	80	

ADDITIONAL MATERIALS

A calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional pages at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question 7.
The quality of written communication will affect the awarding of marks.

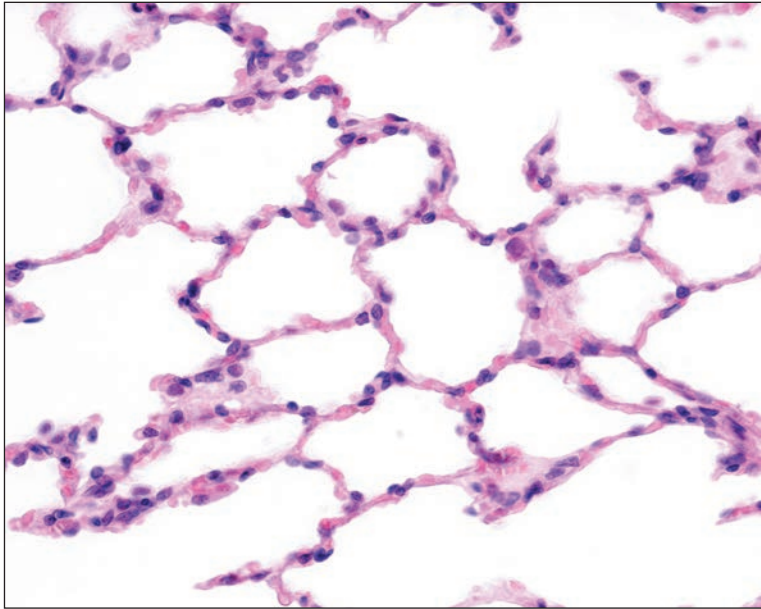


JUN222400U10101

Answer **all** questions.

1. **Image 1.1** shows simple squamous epithelial tissue from the alveoli of the lungs.

Image 1.1



(a) Explain how the shape of the cells is adapted to their function within the tissue. [1]

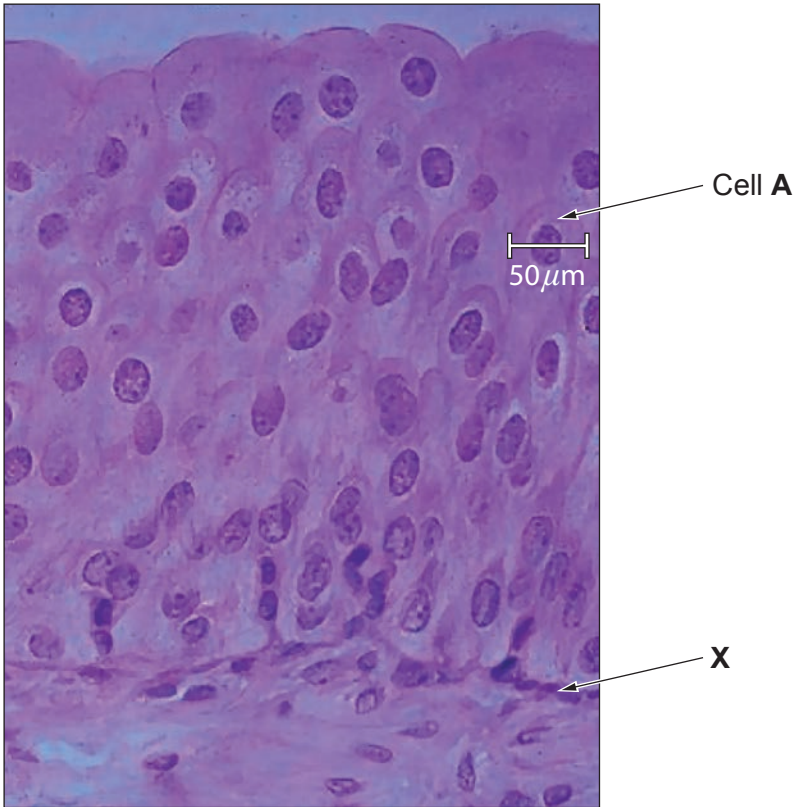
.....

.....



- (b) The urinary bladder is lined with stratified epithelial tissue with a variable number of cell layers. **Image 1.2** shows this tissue when the bladder is relatively empty.

Image 1.2



- (i) Name the structure labelled **X** on **Image 1.2**. [1]
-
- (ii) State why the cells in **Image 1.2** are referred to as a “tissue”. [1]
-
-
- (iii) Explain why some of the cells appear not to have a nucleus in **Image 1.2**. [1]
-
-
- (iv) With reference to **Image 1.1** and **Image 1.2**, describe the main difference between the two types of epithelial tissue. [1]
-
-



Examiner
only

- (c) (i) When the bladder is full, the cells become wider and flatter. Suggest how this property of the epithelial tissue allows the bladder to perform its function. [1]

.....

.....

- (ii) The width of cell **A** in **Image 1.2** is $50\mu\text{m}$. Cell **A** is shown at 42% of its maximum width.
Calculate the maximum width of this cell. **Give your answer to three significant figures.** [2]

Maximum width = μm

- (iii) With reference to the actual width of cell **A**, calculate the magnification of **Image 1.2**. [2]

Magnification = $\times$

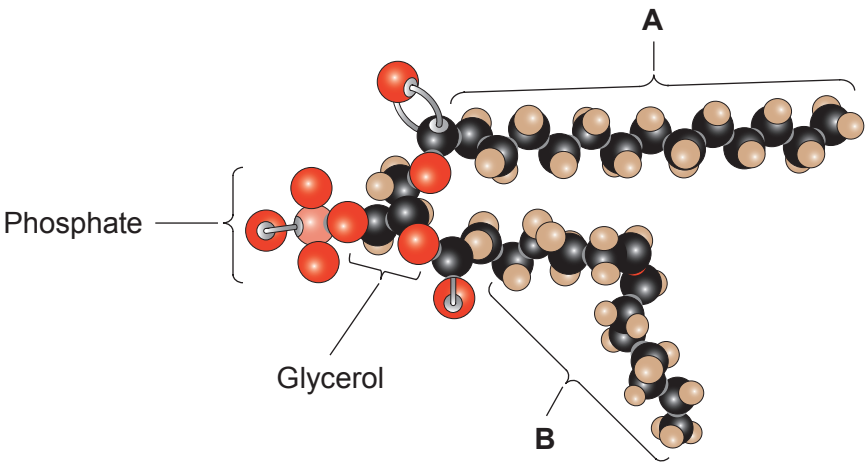
10

2400U101
05



2. (a) **Image 2.1** shows a molecular model of a phospholipid, a component of cell membranes.

Image 2.1



(i) Structures **A** and **B** are both types of fatty acids. Identify both types and explain the difference in their shapes. [2]

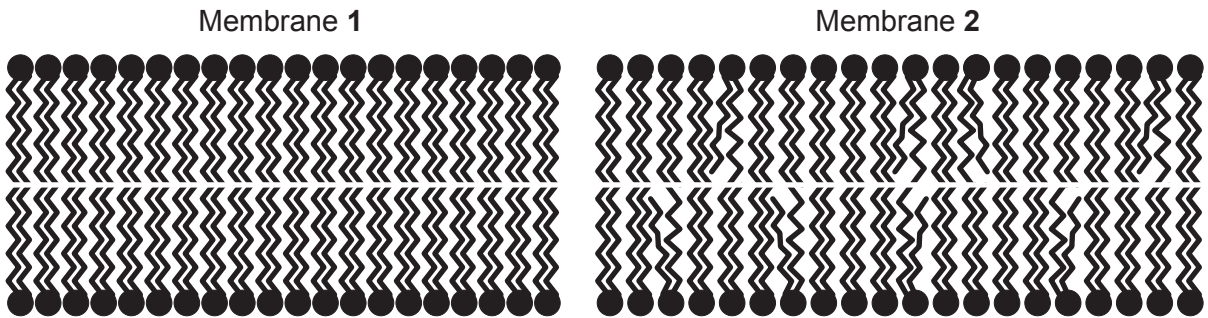
.....

.....

.....

(ii) **Image 2.2** shows two examples of phospholipid bilayers. Membrane 1 has phospholipids containing only type **A** fatty acids and membrane 2 has phospholipids containing both type **A** and type **B**.

Image 2.2



Using the information provided and your own knowledge, explain which membrane you would expect to be more permeable to small molecules. [2]

.....

.....

.....

.....



Examiner
only

(b) (i) Fatty acids are also found as part of triglycerides.

State how triglycerides differ from phospholipids in their structure and in their function in cells.

[2]

.....

.....

.....

.....

.....

(ii) State how the relative proportions of type **A** and **B** fatty acids in the diet can affect human health.

[1]

.....

.....

.....

2400U101
07

7



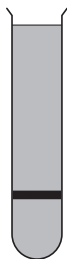
3. (a) In 1958 Meselson and Stahl conducted an investigation into DNA replication.
- Bacteria were grown in a culture medium that contained only the heavy isotope of nitrogen, ^{15}N . After many generations, the bacterial DNA contained only the heavy isotope of nitrogen.
 - Some of the bacteria were then transferred to another culture medium containing only the lighter isotope of nitrogen, ^{14}N .
 - DNA was extracted from the bacteria and centrifuged at the **start** (before cell division), and then after **1, 2 and 3** cell divisions. Heavier DNA was found towards the bottom of the tube after centrifuging.

Image 3.1 shows the distribution of DNA in the tubes at the start and after Division 1 and Division 2. The graphs represent the relative densities of DNA but they are not in the correct order.

(i) **Draw lines** on **Image 3.1** to match up the tubes with the correct graphs. [1]

Image 3.1

Tube



Start

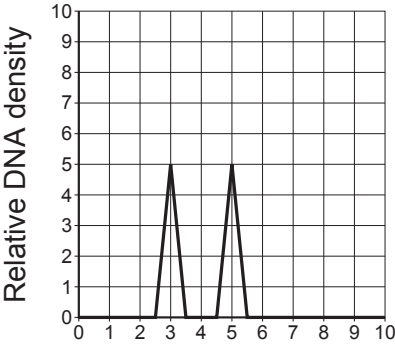


Division 1

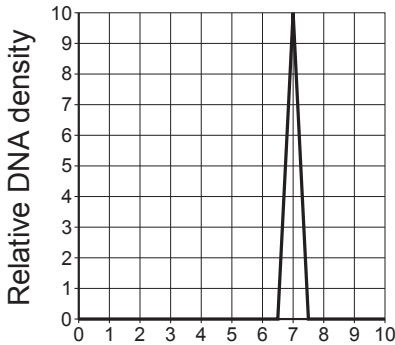


Division 2

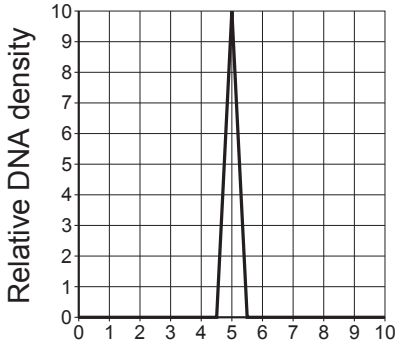
Graph



Top of tube → Bottom of tube



Top of tube → Bottom of tube



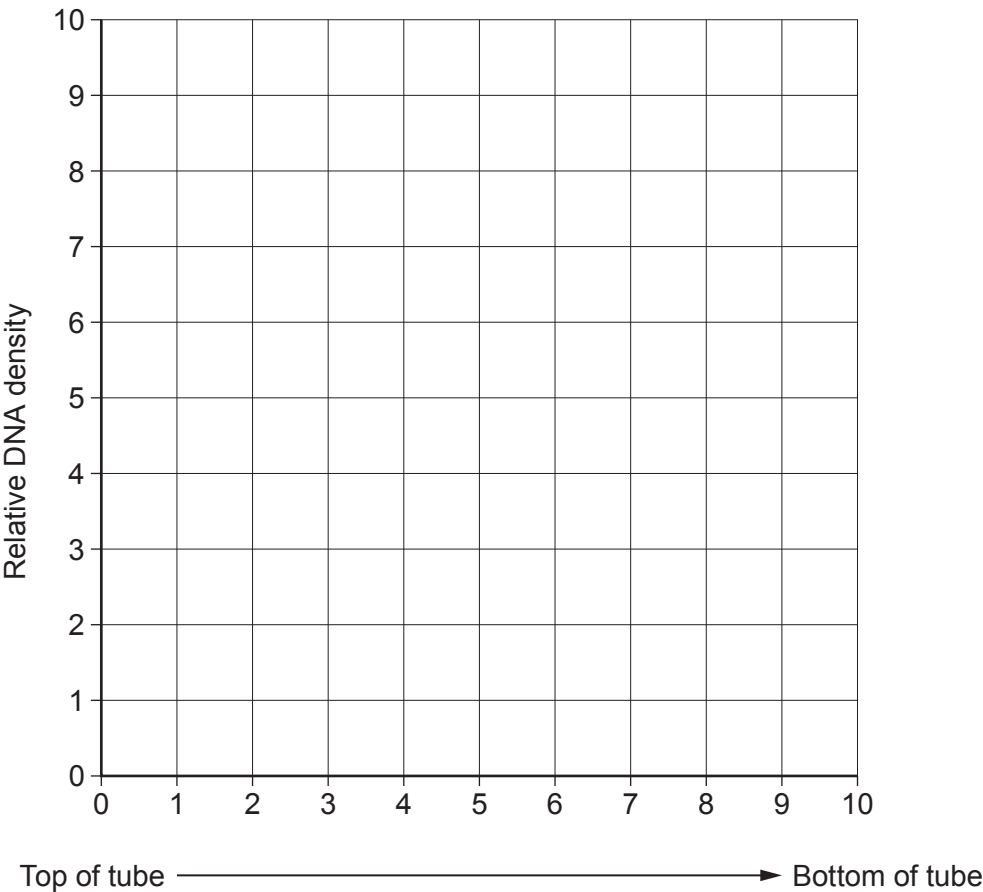
Top of tube → Bottom of tube



(ii) On **Image 3.2** sketch the graph you would expect to see for Division 3.

[3]

Image 3.2

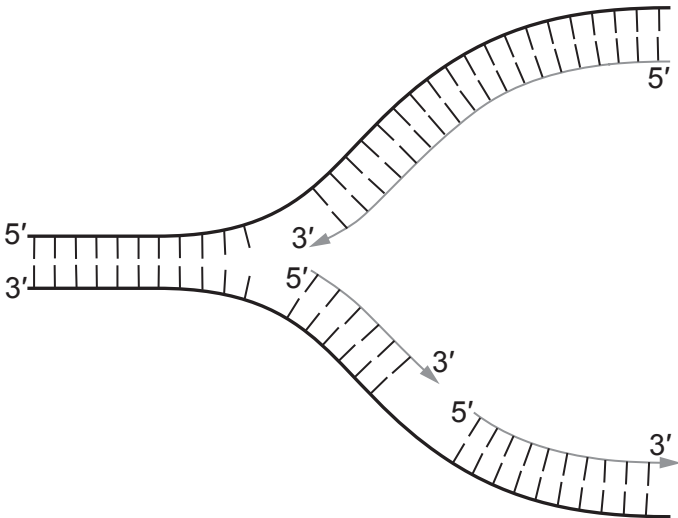


2400U101
09



(b) **Image 3.3** shows a DNA replication fork.

Image 3.3



(i) On **Image 3.3**, mark the position of **one** molecule of DNA polymerase with the letter **X**. [1]

(ii) Describe the role of DNA helicase in this process. [1]

.....

.....

(iii) Explain how **Image 3.3** shows that DNA is replicated by a semi-conservative mechanism. [2]

.....

.....

.....

.....

.....

.....



Examiner
only

- (c) DNA polymerase has a proof-reading function, this means that it will check that it has replicated the DNA correctly and repair it if it is incorrectly synthesised. Use your knowledge of protein synthesis to suggest the importance of the proof-reading role to the organism. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

11

2400U101
11



4. Beetroot contains a red pigment called betalain, which is stored within the cell vacuole. When the cell membranes are disrupted the pigment leaks out into the surrounding solution, colouring it red.

Some students investigated the permeability of the membranes of beetroot cells. A colorimeter was used to measure the absorbance of the surrounding solution. The absorbance of the solution is proportional to the concentration of betalain.

The students made the following prediction:

“The absorbance recorded will be directly proportional to the temperature of incubation.”

The experiment was carried out as follows:

- 6 cubes of fresh beetroot were washed in cold water for 1 minute.
- The cubes were transferred into separate boiling tubes, each containing 20 cm³ of water.
- The boiling tubes were incubated for 10 minutes in beakers at different temperatures which were maintained with the addition of hot or cold water.
- 3 cm³ of water from around the beetroot cubes was removed and a colorimeter was used to measure the absorbance, at 540 nm, of each solution.

The experiment was repeated three times, the results are shown in **Table 4.1**.

Table 4.1

Incubation temperature / °C	Absorbance at 540nm / a.u.			
	Experiment 1	Experiment 2	Experiment 3	Mean
10	12	15	21	16
30	15	20	19	18
40	18	22	32	24
60	51	68	71	63
70	95	96	97	96
80	98	98	98	98

(a) (i) State which temperature provided the most reliable results and use your knowledge of cell structure to explain why this might be the case. [2]

.....

.....

.....

.....

.....

.....

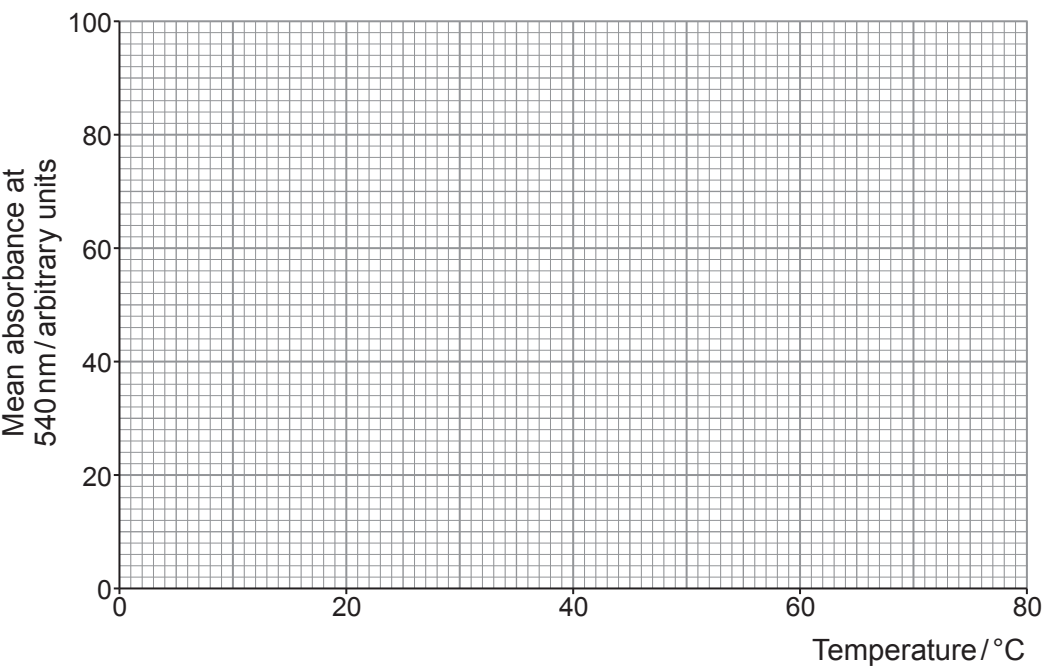


Examiner
only

(ii) Plot the mean results from **Table 4.1** on **Graph 4.2**.

[3]

Graph 4.2



(iii) Explain why the students' prediction was not entirely correct.

[1]

.....

.....

.....

(iv) Use your knowledge of cell membrane structure to explain the shape of **Graph 4.2**.

[2]

.....

.....

.....

.....

.....

.....

.....

.....

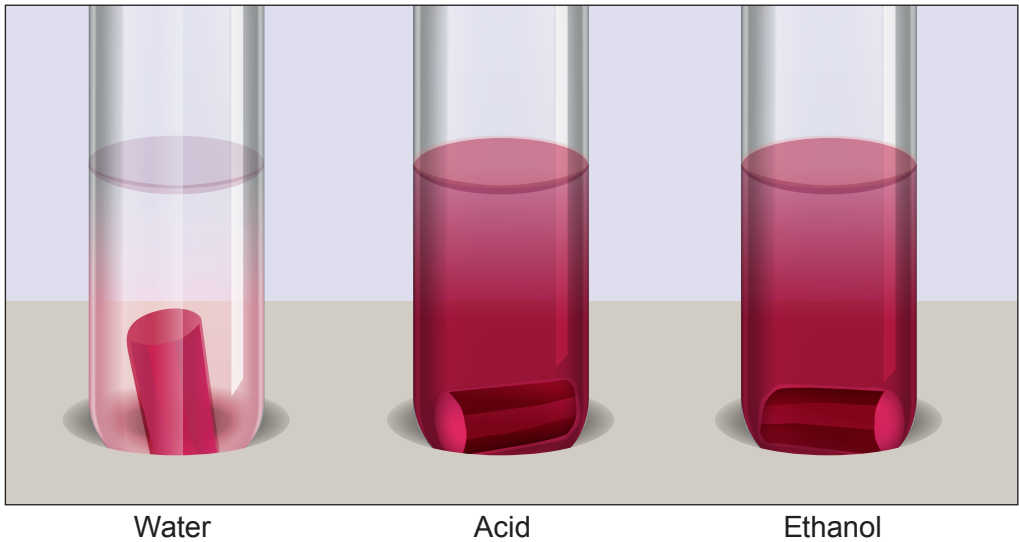
.....

2400U101
13



- (b) Another group of students investigated how the type of surrounding solution affected the permeability of membranes in beetroot. They tested water, acid and ethanol. **Image 4.3** shows the experiment after incubation at 20 °C for 10 minutes.

Image 4.3



- (i) Using your knowledge of the structure of membranes, conclude how both **acid** and **ethanol** affect the membrane and explain the appearance of the solution in these tubes. [3]

.....

.....

.....

.....

.....

.....

- (ii) The students also noted that the beetroot in the water had increased in size. Explain this observation. [3]

.....

.....

.....

.....

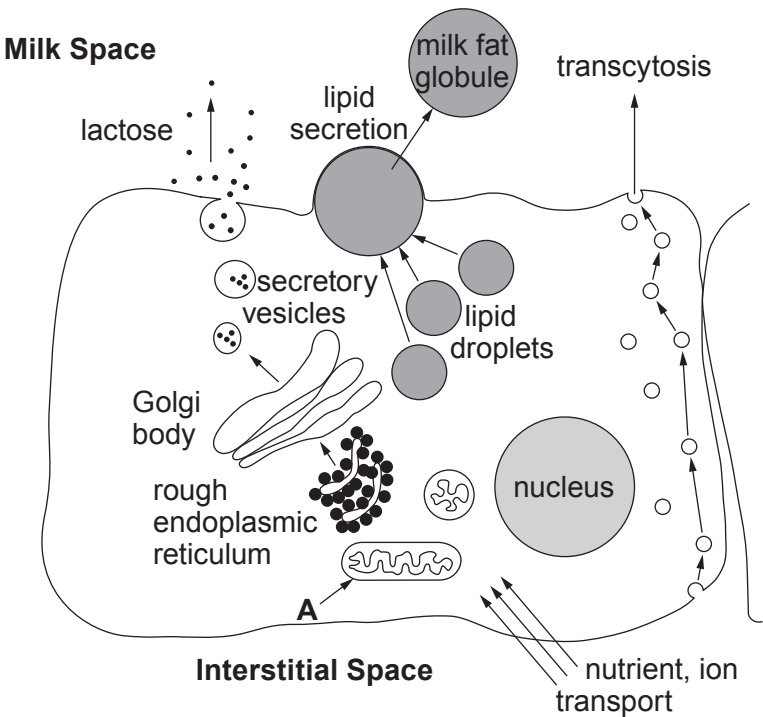
.....

.....



5. Human milk contains all the nutrients a young baby needs in exactly the right proportions. It is formed in the mammary glands by small groups of milk-producing cells in structures called lobules. These cells absorb molecules from the blood and use them to synthesise the lipids, carbohydrates and proteins found in milk. **Image 5.1** shows a milk producing cell.

Image 5.1



Lactose is synthesised in the Golgi body and is transported in vesicles through the cytoplasm. The vesicles increase in diameter as they move towards the plasma membrane. The vesicle membranes fuse with the plasma membrane and the vesicles empty their contents out of the cell. Lipids are also secreted in droplets.

(a) Use the information provided and your own knowledge to answer the following questions.

- (i) State the name of the process by which the vesicles empty their contents out of the milk-producing cell. [1]

.....

- (ii) Identify the organelle labelled **A** and explain its role in the function of this cell. [2]

.....

.....

.....

- (iii) Use **Image 5.1** to suggest how the cell avoids a continual increase in the area of its cell surface membrane. [1]

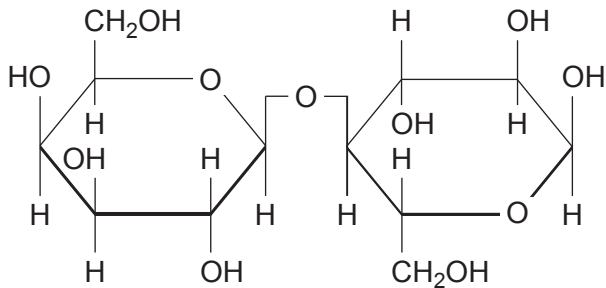
.....

.....



(b) The main carbohydrate in milk is lactose. **Image 5.2** shows a molecule of lactose.

Image 5.2



(i) **Complete Image 5.2** to show the digestion of lactose. [2]

(ii) State:

I. the type of reaction; [1]

.....

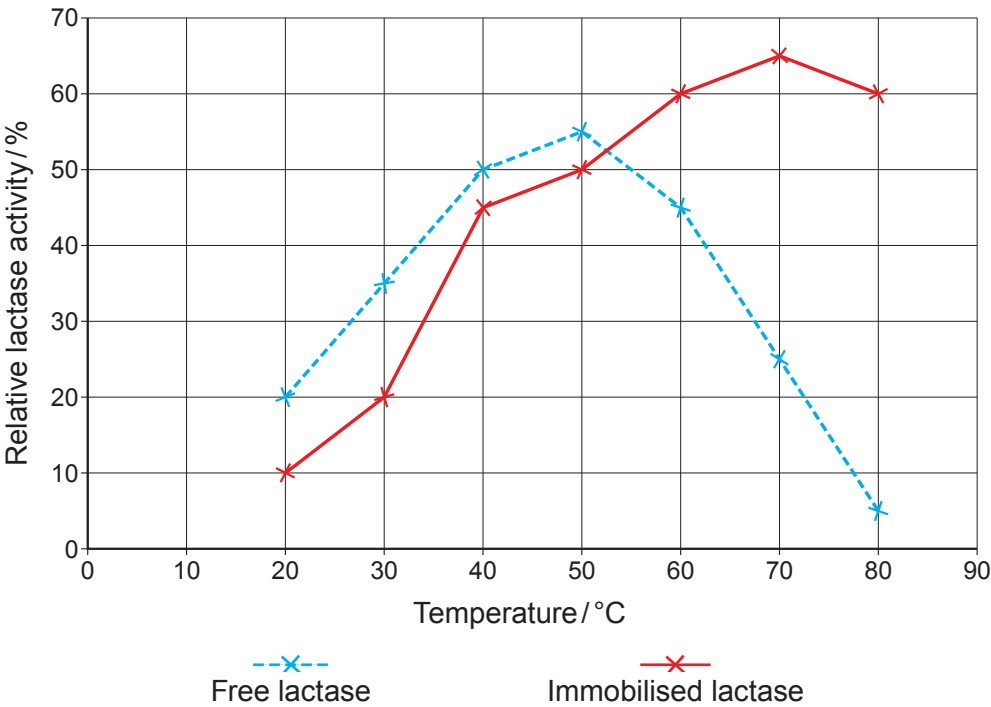
II. the names of the products of digestion of lactose. [1]

.....



- (c) Some people are lactose intolerant and in recent years it has been possible to produce lactose free milk and milk products using the enzyme lactase. One way to do this is to pour the milk through a column of alginate beads containing immobilised lactase. **Graph 5.3** shows the activity of free lactase and immobilised lactase at different temperatures.

Graph 5.3



- (i) Using **Graph 5.3** and your own knowledge, compare the relative lactase activity of the free and immobilised lactase at the following temperatures. Explain each answer.

I. Below 50 °C.

[2]

.....

.....

.....

II. Above 50 °C.

[3]

.....

.....

.....

.....

.....



(ii) Lactose is a reducing sugar. Explain why it is not possible to use Benedict's reagent to monitor the progress of the digestion of lactose by lactase.

[1]

.....

.....

.....

Examiner
only

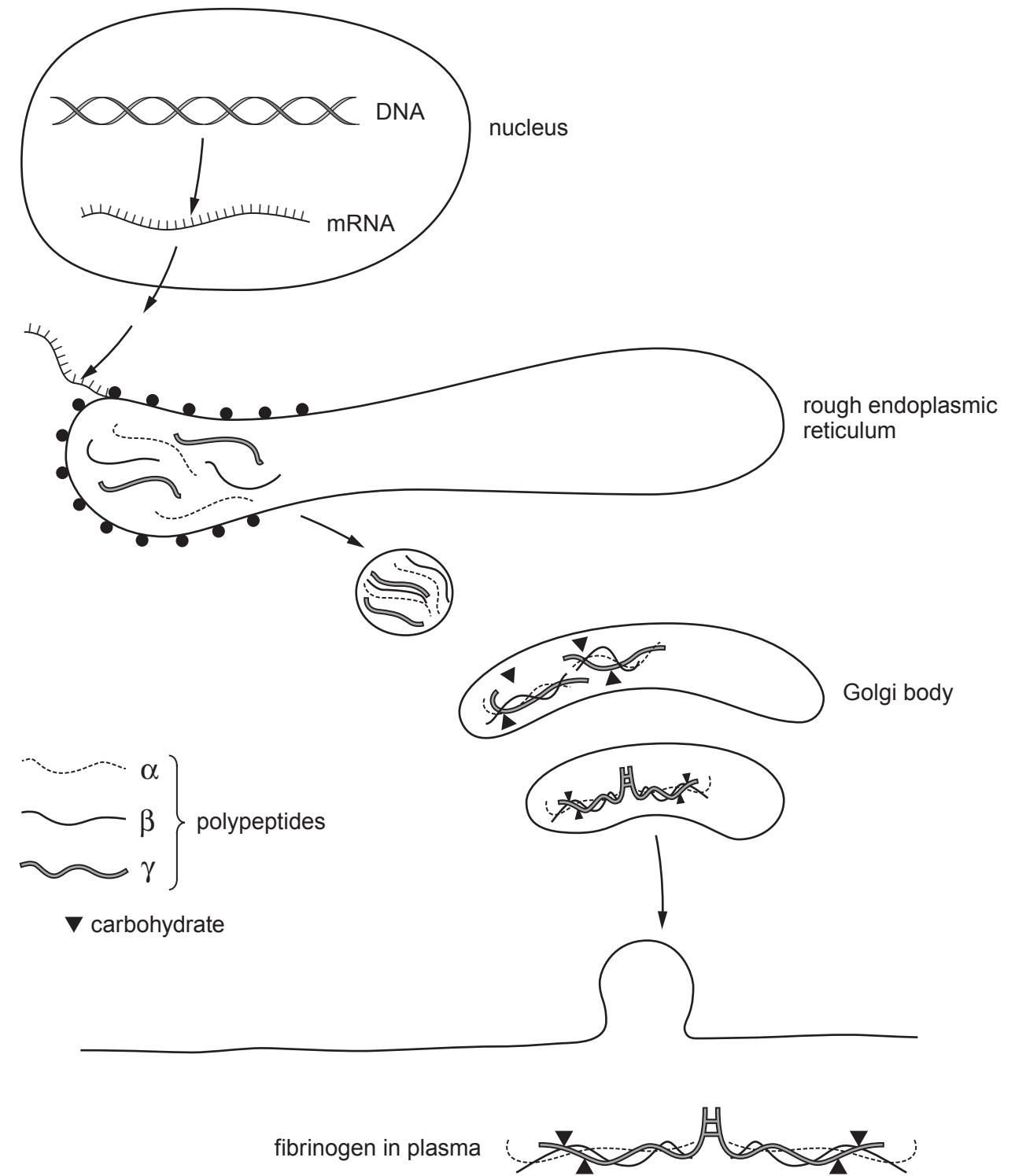
14



6. Fibrinogen is a soluble glycoprotein produced in liver cells and secreted into the plasma. It contains three types of polypeptide chain.

Image 6.1 shows the formation and secretion of fibrinogen.

Image 6.1



Examiner
only

(a) Use **Image 6.1** and your knowledge of proteins and protein synthesis to answer the following questions.

(i) Name the highest level of protein structure shown by fibrinogen. [1]

.....

(ii) Explain the role of the rough endoplasmic reticulum in the production of fibrinogen. [2]

.....
.....
.....
.....

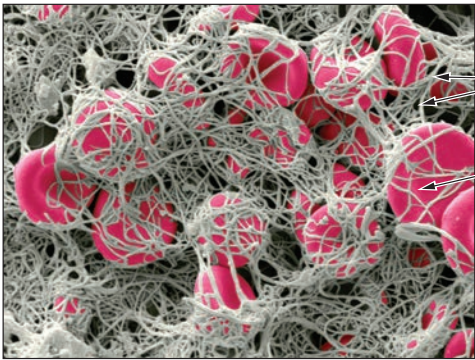
(iii) Suggest the role of the Golgi body in the production of fibrinogen. [2]

.....
.....
.....
.....
.....
.....



- (b) Thrombin is an enzyme which is secreted into blood plasma when tissues are damaged. It catalyses the conversion of fibrinogen, in plasma, into the insoluble, fibrous protein, fibrin. Fibrin seals wounds in tissue. This is shown in **Image 6.2**.

Image 6.2



fibrin

red blood cell

Fibrin is a fibrous protein, fibrinogen is a globular protein.

Using the information provided, explain how the conversion of fibrinogen to fibrin allows the formation of a blood clot. [3]

.....

.....

.....

.....

.....

.....



Examiner
only

(c) Thrombin catalyses the hydrolysis of an arginine-glycine bond in fibrinogen as shown in **Image 6.3**.

Image 6.3

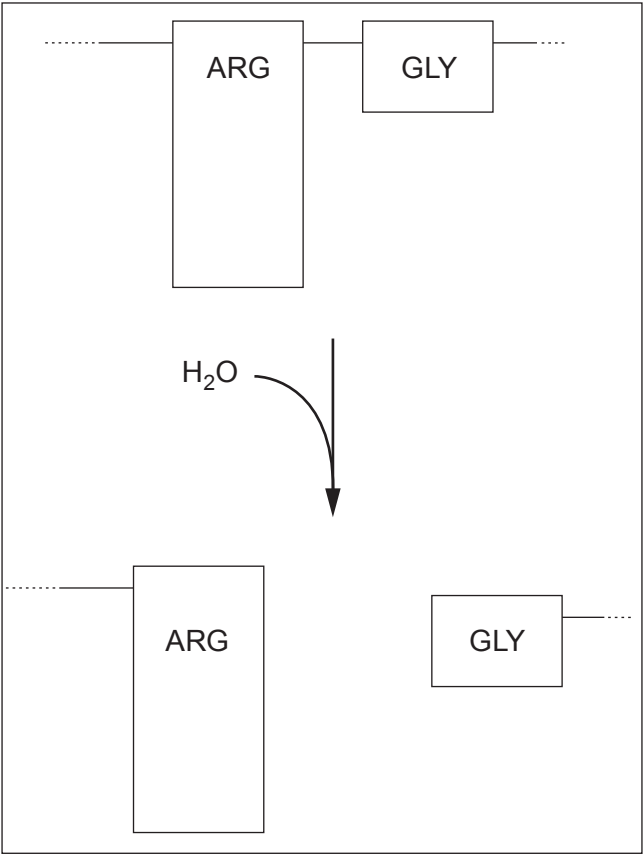
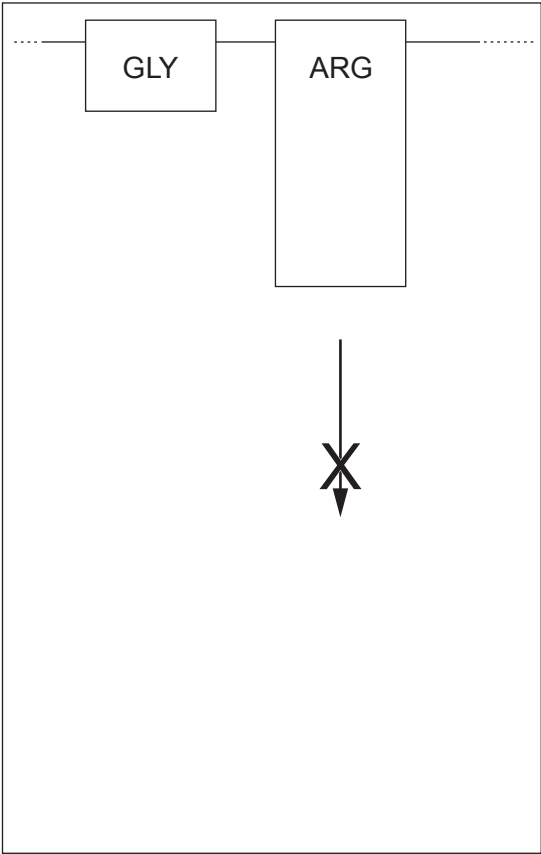


Image 6.4



Name the type of bond broken in **Image 6.3** and explain why thrombin can catalyse this reaction, but it would not catalyse the breaking of a glycine-arginine bond as shown in **Image 6.4**. [2]

.....

.....

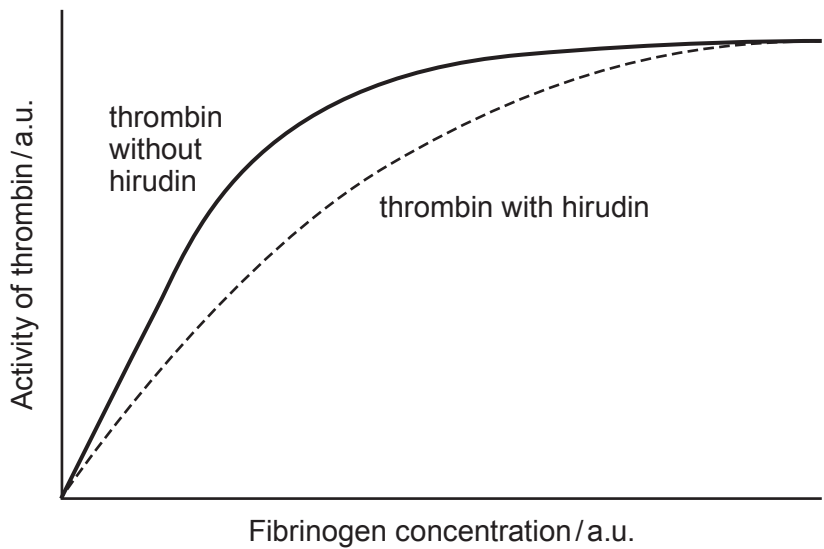
.....



- (d) Leeches are organisms which feed by sucking blood. Their saliva contains many components, including a protein called hirudin.

Graph 6.5 shows the inhibitory effect of hirudin on the activity of thrombin.

Graph 6.5



- (i) Using your knowledge of enzymes, identify and explain the type of inhibition shown by hirudin and suggest how hirudin enables the leech to feed effectively. [4]

.....

.....

.....

.....

.....

.....

.....

.....

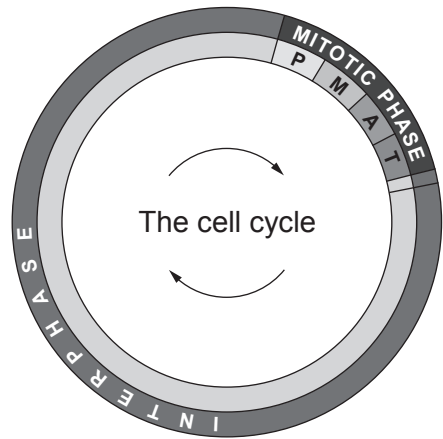
- (ii) Leeches for medical use are farmed in South Wales as a source of hirudin. Suggest a use for hirudin in medicine. [1]

.....



7. A eukaryotic cell cycle is represented in the **Image 7.1**.

Image 7.1



When looking at a group of cells under the microscope, cells undergoing mitosis can be seen as they will have visible chromosomes. The mitotic index, which represents the proportion of cells undergoing mitosis within a group of cells, can be calculated as follows:

$$\text{Mitotic index} = \frac{\text{cells in mitosis}}{\text{total number of cells in field of view}}$$

Cyclins are a family of regulatory proteins that control the progression of the cell cycle. The concentration of cyclin B peaks at the start of mitosis before dropping rapidly as the cells exit mitosis. Some mutations result in abnormally high concentrations of cyclin B.

Describe the events occurring during interphase that are necessary before mitosis can occur.

Using examples, explain why some tissues in humans naturally have a higher mitotic index than others.

Suggest and explain the possible effects of abnormally high concentrations of cyclin B. [9 QER]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....





Examiner
only



Examiner
only

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

END OF PAPER

9





Surname	Centre Number	Candidate Number
First name(s)		2



GCE AS/A LEVEL

2400U10-1



S23-2400U10-1

MONDAY, 15 MAY 2023 – MORNING

BIOLOGY – AS unit 1
Basic Biochemistry and Cell Organisation
1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	12	
2.	13	
3.	10	
4.	11	
5.	13	
6.	12	
7.	9	
Total	80	

ADDITIONAL MATERIALS

A calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional pages at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question 7.
The quality of written communication will affect the awarding of marks.

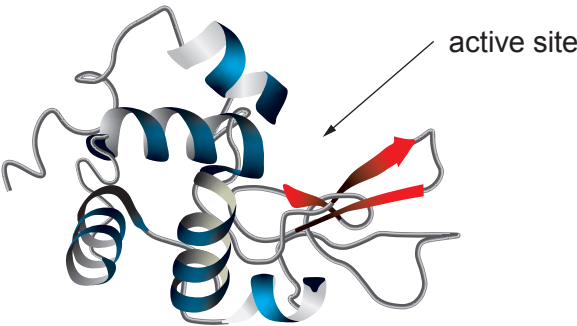


JUN232400U10101

Answer **all** questions.

1. Lysozyme is an enzyme found in saliva and tears. It hydrolyses the carbohydrates in bacterial cell walls. Lysozyme, shown in **Image 1.1**, is a single polypeptide containing 129 amino acids.

Image 1.1 Ribbon diagram of lysozyme



- (a) Lysozyme shows primary, secondary and tertiary structure.

(i) State what is meant by primary structure.

[1]

.....

.....

- (ii) Use the information given and your own knowledge to **complete Table 1.2** to explain how the secondary and tertiary structures of lysozyme are illustrated in **Image 1.1**.

[2]

Table 1.2

Level of structure	Explanation
Secondary	
Tertiary	



Examiner
only

(iii) The cytoplasm of bacterial cells is **hypertonic** to saliva and tears. Suggest how the destruction of the cell wall of bacteria by lysozyme results in the death of the bacteria. [3]

.....

.....

.....

.....

.....

.....

(iv) Lysozyme works by an induced fit mechanism. Explain what this means. [1]

.....

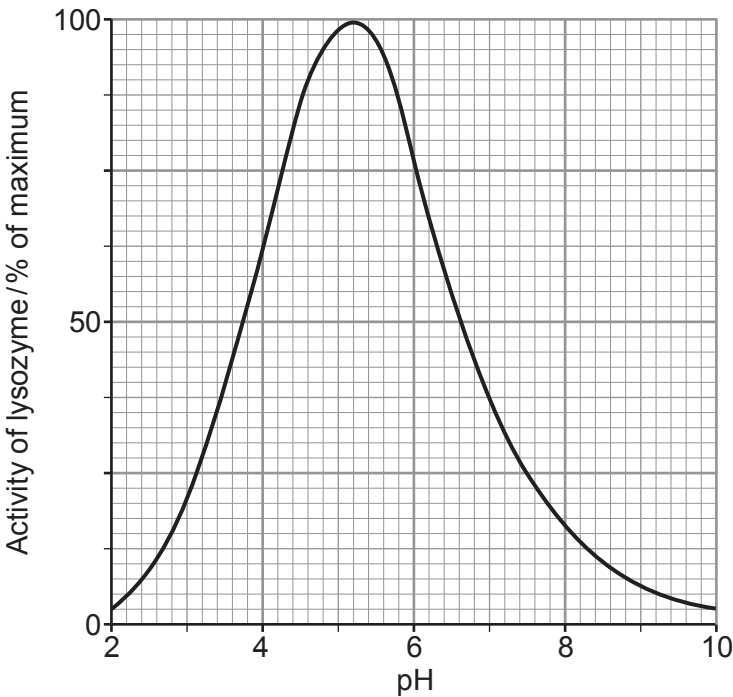
.....

2400U101
03



(b) The effect of pH on the activity of lysozyme is shown in **Graph 1.3**.

Graph 1.3



(i) Describe and explain the effect of pH on the activity of lysozyme. [3]

.....

.....

.....

.....

.....

(ii) Lysozyme has a wider range of pH activity than many other enzymes.

I. Use **Graph 1.3** to find the range of pH over which the activity of lysozyme is more than 50% of the maximum. [1]

Range = to

II. Use the information in the question to suggest **one** advantage of lysozyme having a wider range of pH activity than many other enzymes. [1]

.....

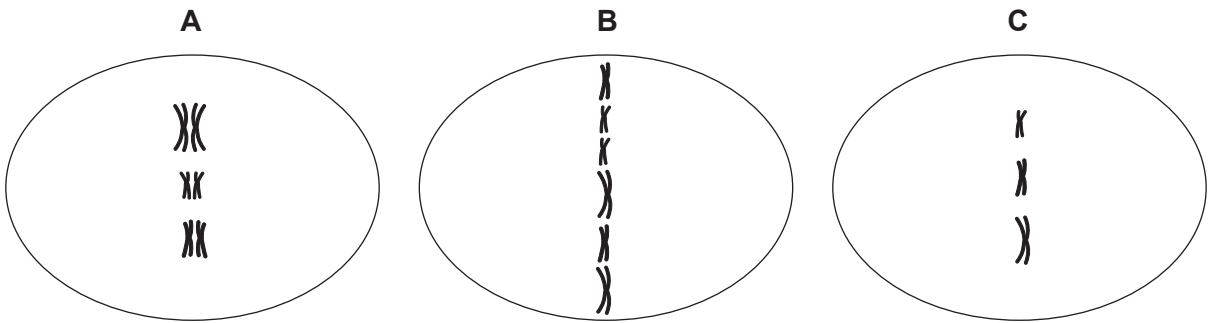
.....



2. **Image 2.1** shows metaphase during cell division of three cells of the mosquito, *Culex pipiens*.

The mosquito has a diploid number of $2n = 6$.

Image 2.1



- (a) (i) Insert the correct letter (**A**, **B** or **C**) into the table to indicate which stage of cell division is represented by each cell. [2]

Metaphase of:	Letter
Mitosis	
Meiosis I	
Meiosis II	

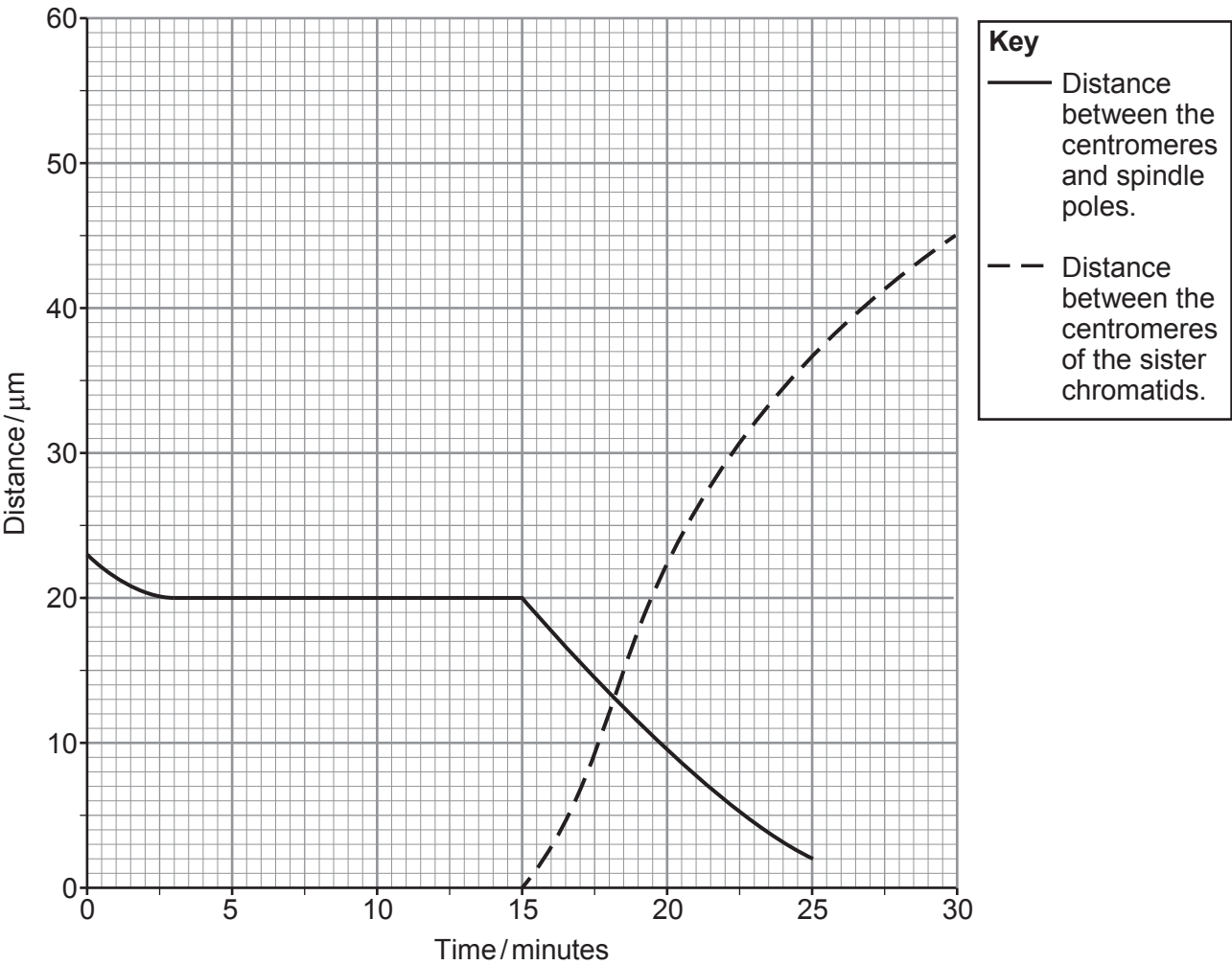
- (ii) State **two** differences in the daughter cells of *Culex pipiens* following mitosis and meiosis. [2]

Difference	Mitosis	Meiosis
1.		
2.		



(b) **Graph 2.2** shows some changes seen in a cell undergoing mitosis.

Graph 2.2



- (i) Calculate the rate of increase in distance between the centromeres of the sister chromatids between 15 and 30 minutes. [2]

Rate = $\mu\text{m minute}^{-1}$

- (ii) Suggest how **Graph 2.2** provides evidence to show what happens to the chromosomes during anaphase of mitosis. [2]

.....

.....

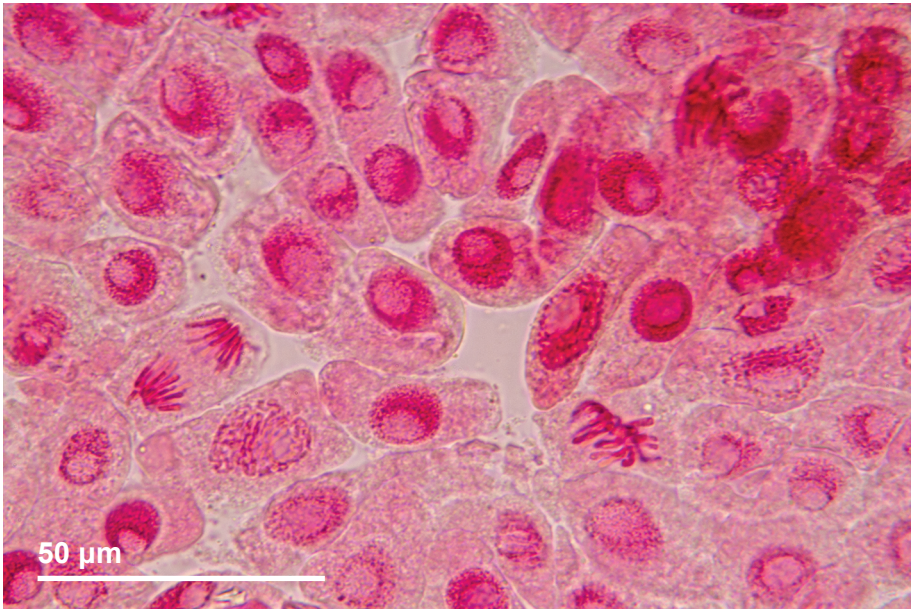
.....

.....



(c) The micrograph in **Image 2.3** shows onion plant cells undergoing mitosis.

Image 2.3



(i) Use the scale bar to calculate the magnification of **Image 2.3**.

[2]

Magnification = $\times$

(ii) On **Image 2.3** label:

[1]

A one cell in anaphase

B one cell in metaphase

(iii) Suggest from which part of the onion plant the cells in **Image 2.3** were taken.
Explain your answer.

[2]

.....

.....

.....

.....

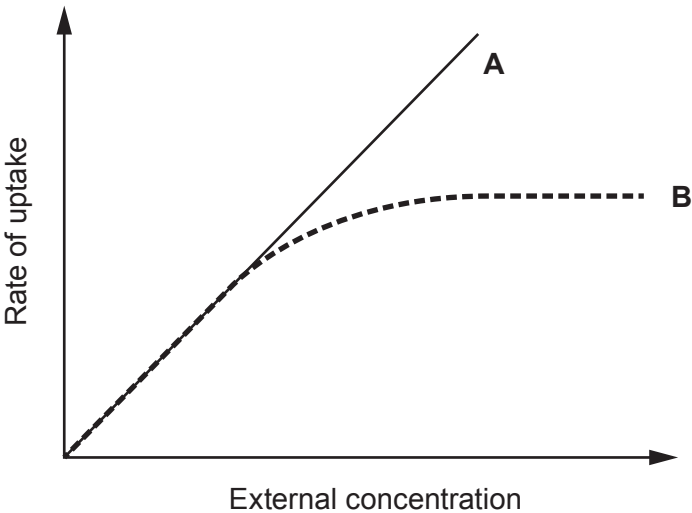


3. (a) Red blood cells need to take in glucose and oxygen from the blood plasma. Glucose is a polar molecule and oxygen is non-polar.

The uptake of glucose and oxygen by red blood cells was measured separately at increasing external concentrations of these two molecules.

Graph 3.1 shows the rate of uptake of glucose and oxygen.

Graph 3.1



- (i) **Complete Table 3.2** to identify:
- which line on **Graph 3.1** represents glucose and which represents oxygen;
 - the type of transport used by each substance.
- [2]

Table 3.2

Line	Substance	Type of transport
A		
B		



Examiner
only

(ii) Explain the difference in the shape of lines **A** and **B** shown on **Graph 3.1**. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(b) Red blood cells are packed full of haemoglobin.
When mature, they do not contain any of the organelles usually found in eukaryotic cells.

Using this information and your knowledge of **cell structure** conclude why mature red blood cells:

(i) cannot make haemoglobin; [2]

.....

.....

.....

.....

(ii) can only transport substances across the cell membrane against a concentration gradient at very low rates. [2]

.....

.....

.....

.....

10

2400U101
09



4. Starch is made up of two molecules, amylose and amylopectin, shown in **Image 4.1**.

Image 4.1

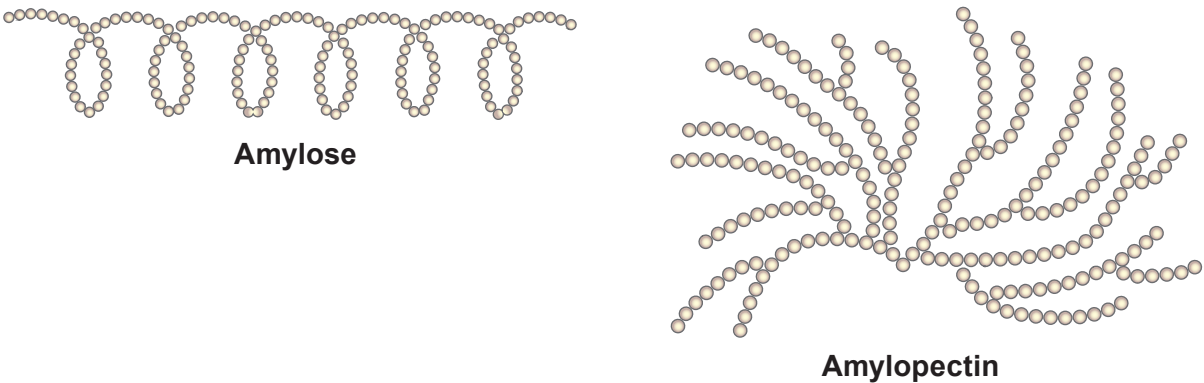
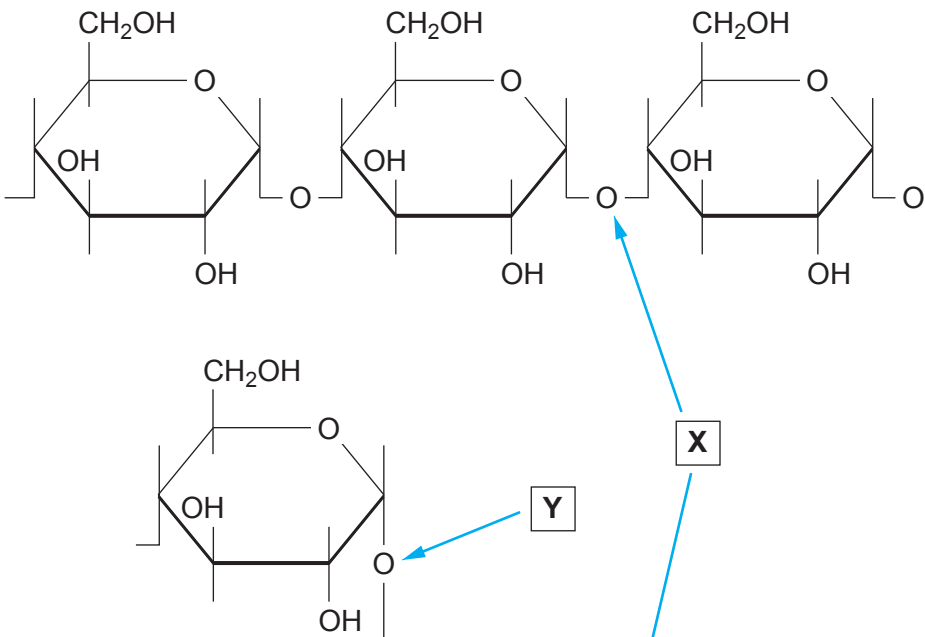


Image 4.2 shows sections of amylose and amylopectin.

Image 4.2

Amylose



(a) (i) State the general name given to **bonds X** and **Y**.

[1]



(ii) State **one** similarity and **one** difference between amylose and amylopectin, shown in **Images 4.1** and **4.2**. [2]

Similarity

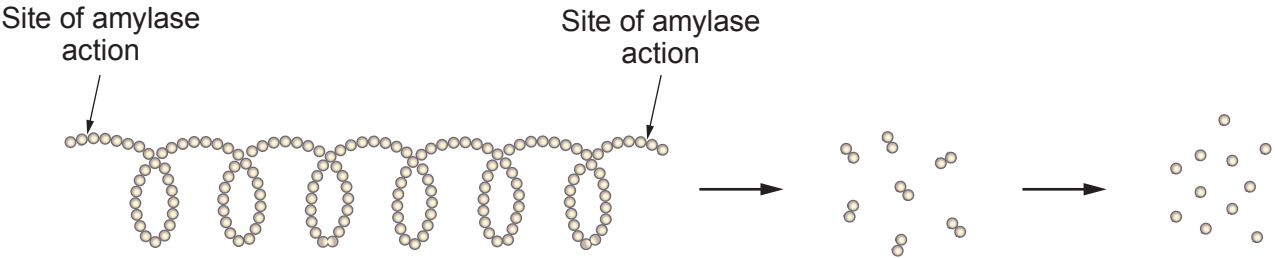
.....

Difference

.....

(b) In order to be absorbed into the blood, starch has to be hydrolysed to produce glucose. The digestion of amylose happens in two stages as shown in **Image 4.3**.

Image 4.3



(i) State what is meant by the term hydrolysis. [1]

.....

.....

Amylopectin is digested in a similar way but needs an additional enzyme, isomaltase, for complete hydrolysis.

(ii) Explain the roles of the **three** enzymes involved in the complete hydrolysis of starch. [3]

.....

.....

.....

.....

.....

.....

.....

.....

2400U101
11



Examiner
only

(iii) One variety of corn used in animal feed has a lower amylose : amylopectin ratio than other varieties. With reference to **Images 4.2** and **4.3**, suggest why this lower ratio results in an increase in the rate of starch hydrolysis. [2]

.....

.....

.....

.....

.....

(iv) Describe a chemical test that could be used to show that starch is no longer present at the end of hydrolysis. [2]

.....

.....

.....

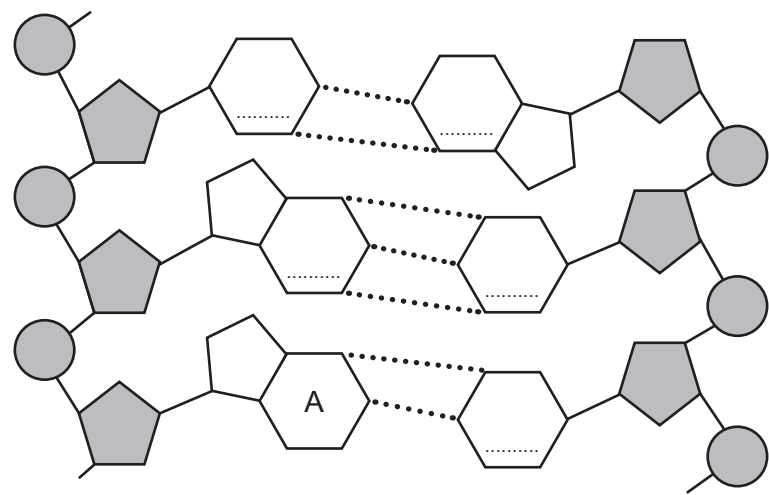
.....

11



5. (a) **Image 5.1** represents part of a DNA molecule.

Image 5.1



- (i) Circle **one** nucleotide in **Image 5.1**. [1]
- (ii) Using the letters **A, C, T, G**, label **all** the organic bases in **Image 5.1**. One adenine is labelled already. [2]
- (iii) In the late 1940s, Erwin Chargaff published his research into the base composition of the DNA in different species. Some of his findings are shown in **Table 5.2**.

Table 5.2

Organism	Percentage of base			
	Adenine	Guanine	Cytosine	Thymine
maize	26.8	22.8	23.2	27.2
chicken	28.0	22.0	21.6	28.4
yeast	31.3	18.7	17.1	32.9
sea urchin	32.8			

I. Complete **Table 5.2** to suggest values for the sea urchin. [1]



Examiner
only

II. Based on the results in **Table 5.2**, state the conclusion that can be made regarding the base composition of DNA. Explain your answer. [3]

.....

.....

.....

.....

.....

.....

(b) Meselson and Stahl also carried out investigations on DNA in the 1950s. They proposed the theory of semi-conservative replication. State what is meant by semi-conservative replication. [2]

.....

.....

.....

.....

(c) DNA contains only four different bases, but proteins can be made of up to 20 different amino acids. Explain how these four bases allow for the production of so many different proteins. [4]

.....

.....

.....

.....

.....

.....

.....

.....

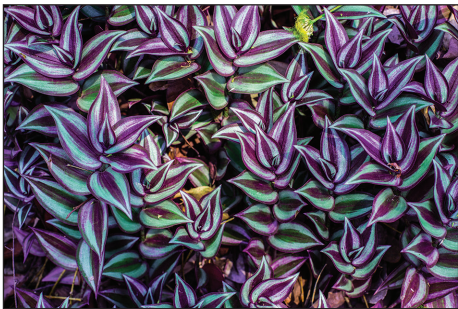
.....

.....



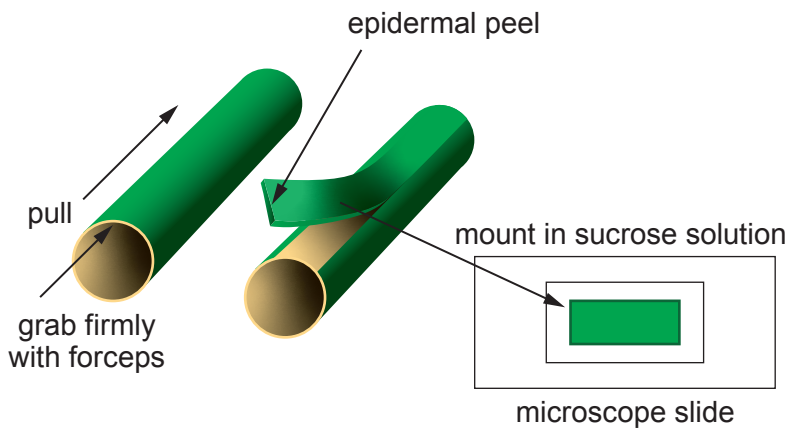
6. *Tradescantia*, shown in **Image 6.1**, is a house plant which is popular due to its ability to survive long periods without watering because its stems can store water.

Image 6.1



The solute potential of the cells of the stem was measured by using epidermal peels as shown in **Image 6.2**. Each peel was mounted on a different microscope slide in a different concentration of sucrose solution.

Image 6.2



After 10 minutes the peels were observed under a microscope. The number of plasmolysed and unplasmolysed cells in each sample was recorded. The results are shown in **Table 6.3**.

Table 6.3

Sucrose concentration/ mol dm ⁻³	Number of plasmolysed cells	Number of unplasmolysed cells	Total cells counted	Percentage of cells plasmolysed
0.0	0	165	165	0.0
0.2	20	174	194	10.3
0.4	90	105	195	46.2
0.6	87	69	156	55.8
0.8	76	54	130	58.4
1.0	172	73		

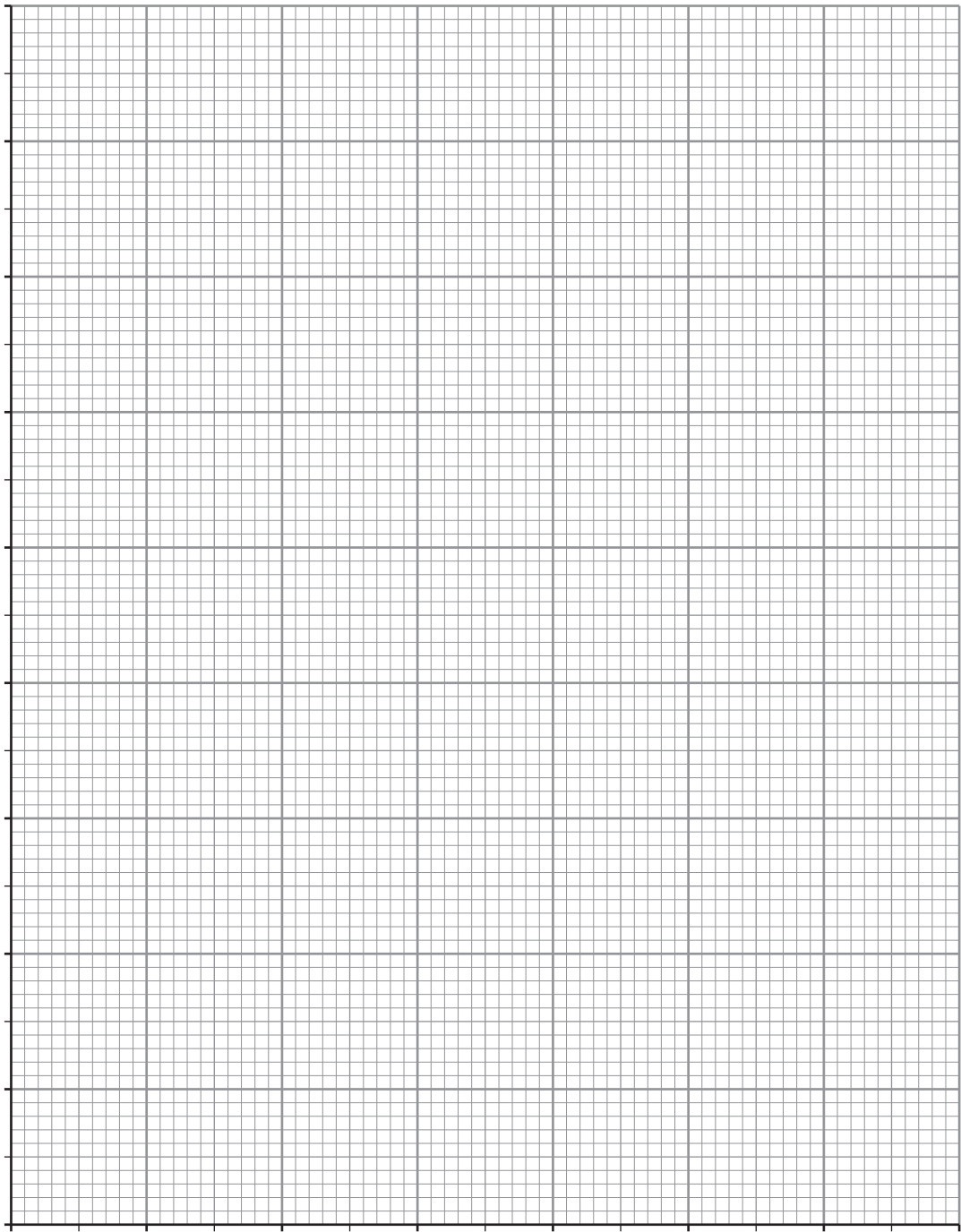
(a) (i) **Complete Table 6.3** for 1.0 mol dm⁻³ sucrose. [2]



(ii) Use the data in **Table 6.3** to draw a graph which shows the percentage of plasmolysed cells against sucrose concentration.

[4]

Examiner
only



Examiner
only

- (iii) I. Use your graph and the formula below to estimate the solute potential of the solution when 50% of the *Tradescantia* cells are plasmolysed. [2]

Solute potential (Ψ_s) = - (sucrose concentration $\times$ 24.35)

Solute potential = kPa

- II. State the pressure potential of the cells in the tissue of *Tradescantia* at this solute concentration and identify the term that describes their condition. [2]

Pressure potential (Ψ_p) = kPa

Term =

- (b) Suggest how the experiment could be altered to improve:

- confidence in the results
- the accuracy of the estimated solute potential in *Tradescantia* cells. [2]

.....

.....

.....

.....

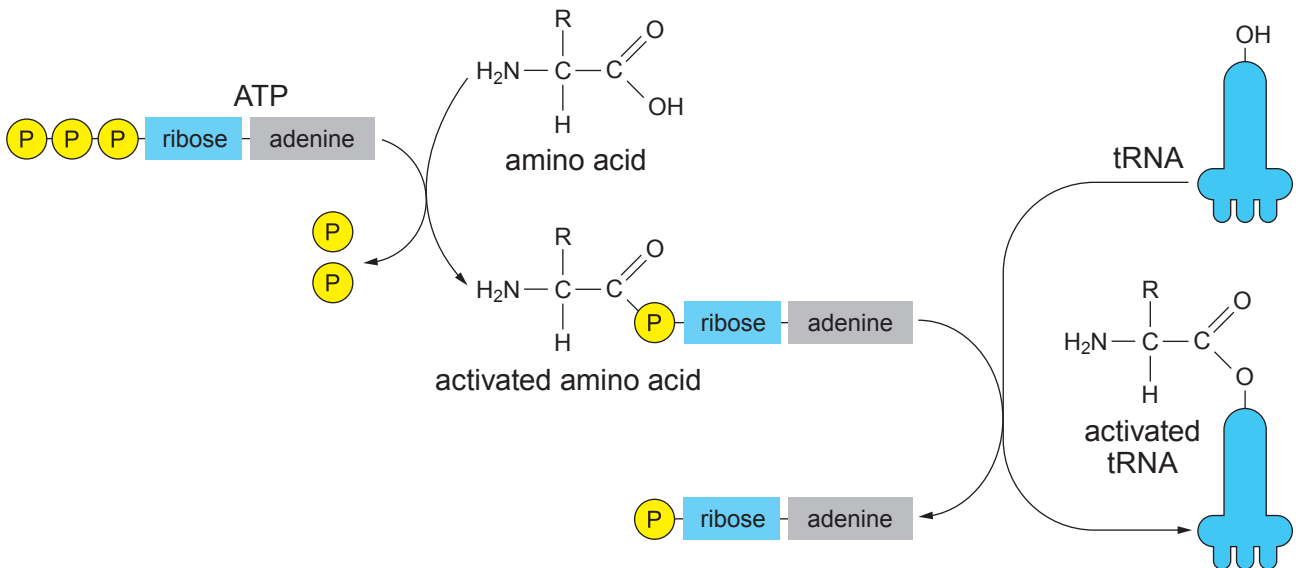
12



Examiner
only

7. DNA, mRNA, rRNA and tRNA are nucleic acids involved in protein synthesis. For tRNA to carry out its role in protein synthesis it needs to undergo the process shown in **Image 7**.

Image 7



Describe the function of each of the **four** types of nucleic acid involved in protein synthesis and state where in the cell each carries out its function. (A detailed description of protein synthesis is not required.)

Using the information given, explain the role of ATP in the process shown in **Image 7**. [9 QER]



Examiner
only

This image shows a single page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

Examiner
only



Examiner
only

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

END OF PAPER

9

